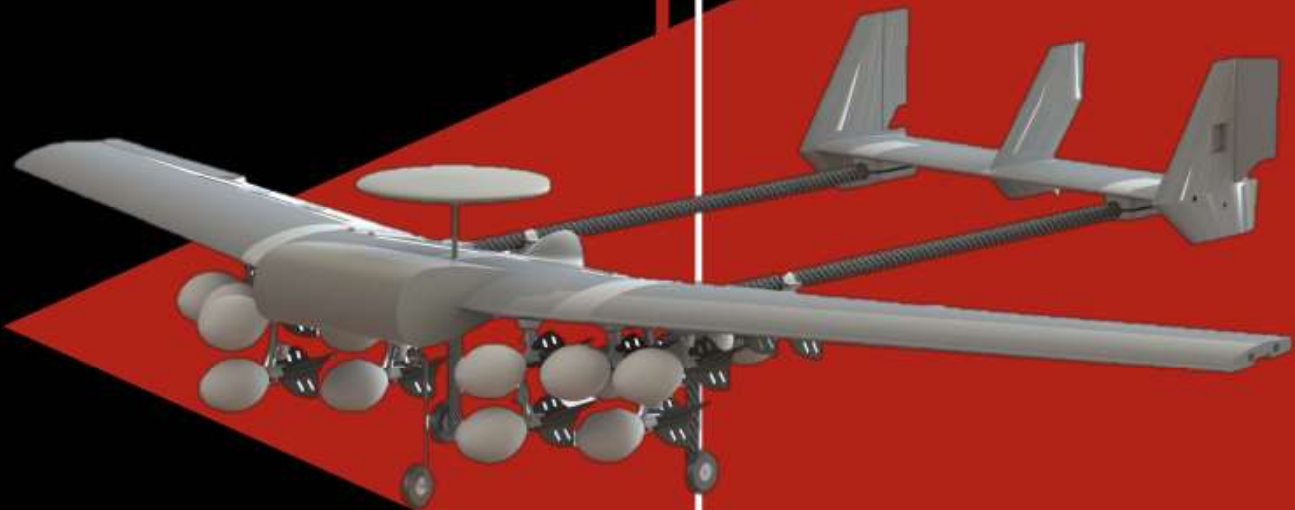




Washington University in St. Louis Design Report



Spirit of St. Louis Woodhams



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Acronyms and Nomenclature

α , AOA	Alpha or angle of attack	ESC	Electronic speed controller
AIAA	American Institute of Aeronautics and Astronautics	FEA	Finite element analysis
AR	Aspect ratio	KM Sim	Kinematic mission simulator
AUW	All up weight (Aircraft weight including payload and passengers)	K_T	Motor torque constant
CAD	Computer-aided design	K_V	Motor velocity constant
CF	Carbon fiber	l_{ac}	Distance from aerodynamic center to the center of gravity
CFD	Computational fluid dynamics	$l_{ac,w}$	Distance from aerodynamic center of the main wing to the center of gravity
CG	Center of gravity	$l_{ac,h}$	Distance from aerodynamic center of the horizontal stabilizer to the center of gravity
CWL	Cubic wing loading	η_h	Dynamic pressure ratio for the horizontal stabilizer
$\bar{c}$	Mean aerodynamic chord	NiCd	Nickel-cadmium
C_D	Coefficient of drag	NiMH	Nickel metal hydride
C_L	Coefficient of lift	PLA	Polylactic acid
C_{L0}	Coefficient of lift at 0 AOA	RPM	Rotations per minute
$C_{L,\alpha}$	Coefficient of lift with respect to alpha	Rx	Receiver
$C_{L,\alpha,w}$	Coefficient of lift with respect to alpha on the main wing	STOL	Short takeoff and landing
$C_{L,\alpha,h}$	Coefficient of lift with respect to alpha on the horizontal stabilizer	SU	Student Union
$C_{l,\beta}$	Coefficient of roll with respect to beta	T/W	Thrust-to-weight ratio
$C_{n,\beta}$	Directional stability coefficient	V_∞	Freestream velocity
$C_{m,\alpha}$	Longitudinal stability coefficient	ν	Kinematic viscosity
$\frac{dc}{d\alpha}$	Washout factor affecting airflow over the horizontal stabilizer with respect to alpha	$V_{logic, min}$	Minimum signal voltage for smooth servo operation
DBF	Design/Build/Fly	$V_{logic, abs min}$	Minimum signal voltage for any servo operation
EPRB	Engineering Project Review Board	WU	Washington University in St. Louis
EPS	Expanded Polystyrene	WUDBF	Washington University Design/Build/Fly

1. Executive Summary

1.1 Overview

This report describes the design, testing, and manufacturing processes of the *Spirit of St. Louis Woodhams*, or simply the *Spirit*. Washington University in St. Louis Design-Build-Fly (WUDBF) team's entry in the American Institute of Aeronautics and Astronautics (AIAA) 2019 Design-Build-Fly competition in Tucson, AZ. WUDBF's goal is to design an aircraft to compete in this competition and place within the top ten teams. By the conclusion of several design iterations, WUDBF created the *Spirit* to meet and exceed the general and mission-specific requirements. WUDBF will be able to achieve a high score in the DBF competition because of the organization's strict timeline, workload delegation, weighted design considerations, and commitment to excellently designed aircraft.

1.2 Approach

WUDBF will succeed at the competition by properly managing its resources of time, money, and workforce. WUDBF has included redundant safety factors in its schedule by allocating time for flight testing of the competition vehicle, ensuring the team has a sufficient budget (with minimum funding of \$25,000) and assigning discrete roles to each member on the roster. From a design perspective, WUDBF will analyze the mission parameters and scoring criteria through a tradespace study (Fig. 3.5) to discover which configurations score highest, including the *Greased Lightning*, the *Pusher-Puller* and the *Spirit* (Fig. 3.4). These aircraft had design features that were key to their projected performance (Section 1.3).

1.3 Summary of Design

Analysis of the scoring guidelines suggests that in this year's competition the score is primarily determined by Mission 3 capabilities. With this in mind, WUDBF's tradespace analysis suggests a fast, efficient plane with high carrying capacity and short take-off capability will score the highest at the competition. The scoring system strongly favors aircraft that can yield a large number of scoring laps in Mission 3, as quantified in Section 3.3. Thus, the *Spirit* design will maximize flight speed, carrying capacity, and efficiency while minimizing take-off distance. To do this, WUDBF implemented a single-engine pusher similar to the *F-35B*. The *Spirit* has a tapered swept wing with an H-tail and trailing edge flaps configuration. This configuration allows the same static stability and performance as a traditional aircraft and short takeoff capabilities due to thrust vectoring. This design was chosen because it demonstrated a high scoring capacity and limited risk (Fig. 3.5).

1.4 Key Mission Requirements and Associated Design Features

This year's competition asks teams to build an aircraft based on "aircraft carrier operations," capable of taking off in a relatively short distance and folding its wings for a stowed configuration. To most accurately simulate the performance of an aircraft that combines the capabilities of these two classes of aircraft, the rules outline various general requirements. These requirements include a wingspan of at least 4 ft, a limited stowed configuration size that must fit within a 3 ft wide and 2 ft high box, and a remote

transition from its stowed to its flight configuration. Additionally, the nose and all landing gear must fit within a depth of 2 ft, the aircraft must remotely lock itself in its flight configuration, and it must take off a ramp that is 4 ft wide, 10 ft long, and at an angle of 5° above the runway. During takeoff, the aircraft cannot touch back down on the runway after the ramp and it must have a tailhook that can be held by a member of the ground crew and be secured at the before takeoff. The tailhook must be beneath the aircraft, on the fuselage centerline. While in flight, the aircraft must be able to carry and drop a minimum of 4 attack stores, all of which must be mounted under the wing. Any additional attack stores can be mounted under the wing or beneath the fuselage. All of these stores must remotely detach individually and independently and have 1/2 in of clearance between each other and other non-release mechanism structures. Finally, the aircraft must carry a removable, 12 in diameter, 1 in thick radome that can be remotely controlled to rotate while in flight. It must be mounted 3 in from any non-mounting structure.

During the timed ground mission, the team must prove the aircraft can be remotely commanded into flight configuration, the radome is removable, and this component can also be remotely controlled. Additionally, a team member must mount four attack stores on the aircraft. Then the pilot must demonstrate all flight control capabilities and perform a propulsion systems check.

1.5 Aircraft Solution Performance Summary

WUDBF has designed the *Spirit* with all the previously mentioned design requirements in mind. Table 1.1 provides basic information about the aircraft's design and projected performance based on aircraft CAD models and a kinematic simulator (KM Sim).

Table 1.1 Basic aircraft information and performance data.

Empty Weight (kg)	6.50	Cruise Speed (m/s)	25.0
Takeoff Weight (kg)	8.10	Wingspan (m)	2.50
Takeoff Distance (m)	3.05	Planform Area (m ²)	1.17
Top Speed (m/s)	33.0	Planform Loading (Pa)	340
Number of Stores	16	Estimated Score	20-21

2. Management Summary

2.1 Organization

To facilitate efficient communication and work, WUDBF is organized into an Executive Board, Advisors, and five Design Divisions. Fig. 2.1 presents the hierarchy of the team. The Executive Board consists of a Technical President, an Administrative President, a Vice President of Communications, and a Vice President of Finance. Executive Board members are responsible for aircraft technical decisions, administrative tasks, resource allocation, and public relations. The Advisors include

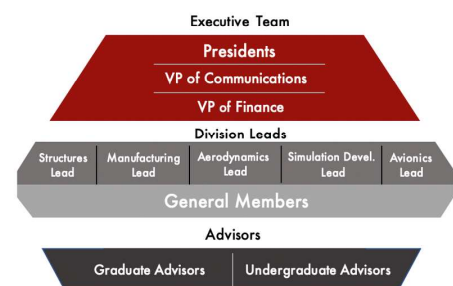


Fig. 2.1 WUDBF hierarchy.

Graduate Advisors who provide industry and/or graduate level expertise and guidance to the team. The Undergraduate Advisors are defined as any student who was previously a division lead or on the executive team and they counsel current team leads. There are five divisions (described in Table 2.1) and each is essential to the design and construction of the *Spirit*.

Table 2.1 Relevant roles, assignments, and skills of WUDBF division personnel.

	Aerodynamics	Structures	Manufacturing	Avionics	Simulation Devel.
Assignments	- Outer mold line of the aircraft - Performance goals met - Stability & Control	- Airframe design - Structural analysis - Mass properties	- Create manufacturable drawings - Manufacture from CAD - Material selection	- Design/optimization of propulsion systems - Control surface integration - Radio control & telemetry	- Aircraft & mission development for the current year's rules and objectives - Aircraft model design in RealFlight for pilot training
Skills	- Aircraft stability, control, optimization - Numerical & CFD analysis	- CAD - Structural FEA - Mass properties analysis	- Laser cutting, 3D printing, machining - Reading technical drawings	- Propulsion theory - Circuit design - Embedded software - Soldering & crimping	- MATLAB & Simulink - Aircraft physics - Modeling

WUDBF follows a strict schedule, as shown in Fig. 2.1, with the goal of completing deadlines on time or early. WUDBF managed to stay on or ahead of schedule during critical design stages and was able to adapt to manufacturing and testing challenges to complete the *Spirit* on time.

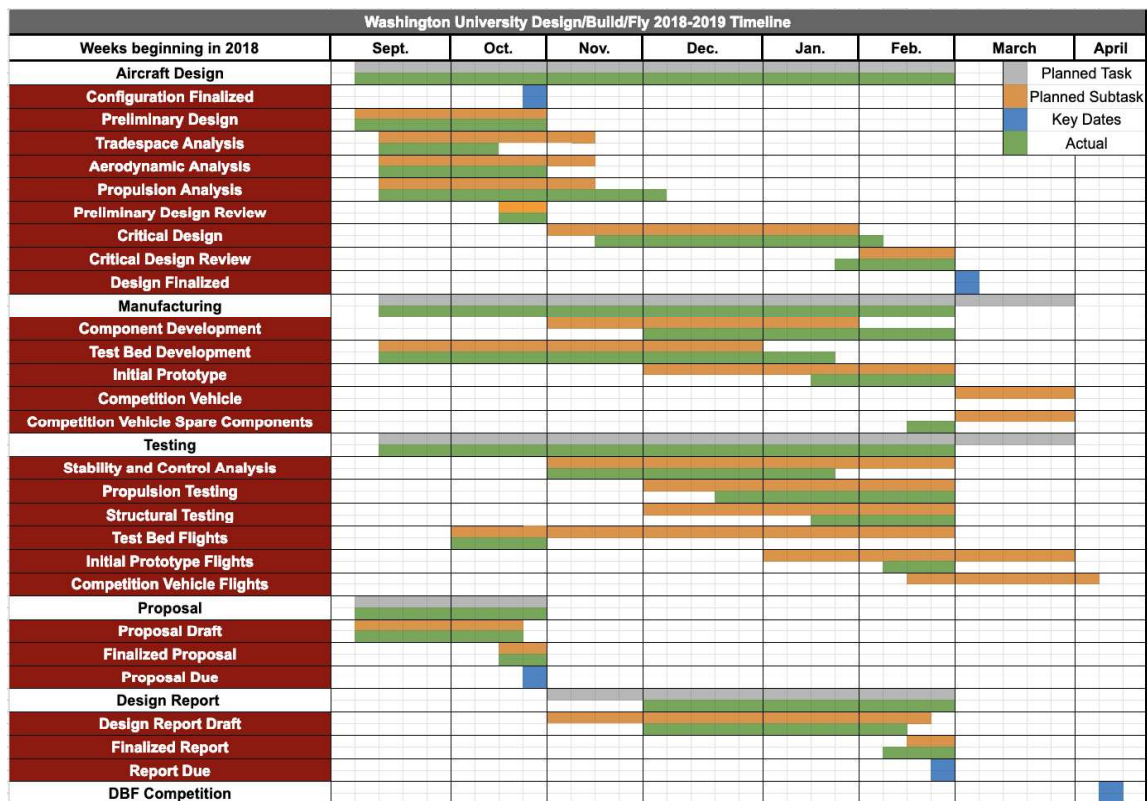


Fig. 2.1 Gantt chart of WUDBF's design and build timeline for the *Spirit*.

3. Conceptual Design

3.1 Mission Requirements

For the 2018-2019 DBF competition, Mission 1, Mission 2, Mission 3, and the Ground Mission will be completed in accordance with the AIAA's rules and regulations. A sample mission path is detailed in Fig. 3.1.

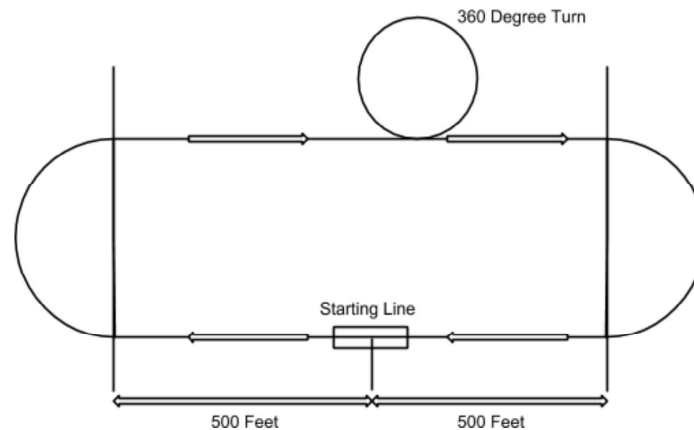


Fig. 3.1 AIAA 2019 DBF competition course layout.

3.1.1 Mission 1

Mission 1 is the delivery mission. The vehicle will begin this mission in its stowed configuration. Within five minutes of entering the staging box, the wings will be remotely transitioned to their flight configurations and remotely locked. After a wing tip test to check for successful wing locking, the plane will be moved to the takeoff ramp. There, a crew member will insert the vehicle's shunt plug, thereby arming the propulsion system, and manually restrain the vehicle via its tailhook as the pilot powers up its propeller. At the pilot's verbal cue, the crew member will release the tailhook and the vehicle will take off into the air. After taking off, the vehicle has five minutes to complete three laps of the course. After three laps, the pilot will successfully land the aircraft.

3.1.2 Mission 2

Mission 2 is the reconnaissance mission. As in Mission 1, the vehicle will enter the staging box without a payload and with its wings stowed. During the five-minute staging period, the radome will be manually installed, the wings will remotely transition to their flight configuration and lock, and the wing tip locking feature will be checked. The vehicle will move to the take-off ramp and a flight crew member will restrain the vehicle by its tailhook as the propulsion system activates. At the pilot's verbal command, the ground crew member will release the tailhook, allowing the vehicle to take off and complete the mission. Within five minutes of the tailhook release, the vehicle will complete three laps. After the first 180° turn, the pilot will command the radome to begin rotating, and it will stop turning after the second 180° turn is accomplished. After a total of three laps, the pilot will execute a successful landing.

3.1.3 Mission 3

Mission 3 is the attack mission. As with Missions 1 and 2, the vehicle will initiate the mission in its stowed configuration. Once inside the staging box, the number of attack stores approved during the technical inspection will be supplied and manually installed during the five-minute staging window. Additionally, during the staging time, the wings will remotely transition to flight configuration, lock, and the wing tip locking feature will be checked. After staging is complete, the vehicle will taxi to the takeoff ramp. While a flight crew member restrains the vehicle by its tailhook, the propulsion system will power up and at the pilot's verbal command, the crew member will release the hook, which enables takeoff and begins the mission timing. After the vehicle is airborne and the first upwind 180° turn is complete, the pilot will remotely command the first store to be dropped during the downwind portion of the flight path. Ten minutes are allotted for this mission, and after all laps and store drops are complete, the vehicle will successfully land.

3.1.4 Ground Mission

The ground mission is timed and must be completed prior to Missions 2 and 3. The vehicle, in its stowed configuration, will initially be in a ten foot by ten foot "mission box" with four attack stores and the uninstalled radome. When mission officials give a verbal, "GO," the pilot will remotely switch the aircraft to flight configuration and the assembly crew members will cross the starting line, run to the aircraft to install the radome, and then back to cross the start/finish line, at which point time will pause. After the pilot proves the radome rotation can be controlled remotely, a ground mission official will issue another "GO," at which point time will resume, crew members will cross the starting line, uninstall the radome, position four attack stores on the aircraft, and return to the pilot behind the start/stop line. Time will be stopped again while the pilot performs all flight controls, including a demonstration of the propulsion systems. To complete the mission, an assembly crew member will disarm the aircraft and the pilot will show that each store can be released individually.

3.2 Design Philosophy

In keeping with past years, the 2019 DBF competition has defined its mission requirements based on the roles of real-world aircraft. This year, the classes of aircraft being considered are those who take off from amphibious assault ships (STOL capable vehicles) and those that provide early warning detection. Therefore this year's design must be capable of emulating the qualities of STOL and Range and Velocity. These capabilities are tested during the four missions.

3.3 Design Requirements

Aircraft with the highest speed, range, and payload capacity that can still perform the required short take off will perform the best and earn the most points in the competition. The vast majority of the possible points will be earned in Mission 3.

Mission 1 requires the aircraft is stable, controllable, can take off in the confined takeoff distance, and meets minimum range requirements. It represents basic mission requirements that must be exceeded to perform optimally during Missions 2 and 3.

Mission 2 stresses aircraft speed. To score well in the time trial, the aircraft must maximize cruise velocity; however, to take off from the ramp, the aircraft must generate considerable lift at low velocities. Takeoff therefore requires a high lift setup and cruise performance requires a low drag setup.

Mission 3 scores the most points. This mission adds range and payload considerations to the requirements of the previous missions. Maximizing the release of store payloads maximizes score but considerably increases takeoff weight, requiring high lift devices and a high thrust to weight ratio (T/W). This mission also requires the aircraft to have a large enough range to remain airborne for the full ten-minute window to release all stores. During this time window, the aircraft must maintain a high cruise velocity so the laps can be completed quickly enough to drop all of the stores.

A tradeoff must be made between takeoff performance and flight range and velocity to maximize the Mission 3 score. An aircraft fully optimized for in-flight performance has high wing loading and short, stubby wings that increase its velocity and range due to decreased drag. However, this same aircraft requires a very high takeoff velocity and cannot take off in the allotted 10 ft. An aircraft optimized for takeoff, on the other hand, has a very large wing with low wing loading. This aircraft has more viscous drag. This drag limits mission range to fewer laps in the allotted time.

Since Mission 3 has the highest point potential and all requirements for Missions 1 and 2 are met by designing for Mission 3, WUDBF designed the *Spirit* to meet this mission's requirements and maximize Mission 3 score.

While the Ground Mission does not have aerodynamic and stability performance requirements, it does add geometrical constraints. The two-foot height limit both reduces the possible vertical stabilizer height and limits landing gear height. The three-foot folded wing constraint limits fuselage and wing size. Since the stores must be stored externally, the size of the fuselage is far smaller than the constraints. However, the three-foot box significantly constraints wing shape.

3.4 Configuration Selection

The four different wing folding techniques considered were STO-wing (a wing that folds along the side of the aircraft fuselage), folding wing, rotating wing, and a hybrid folding and rotating wing. Traditionally defined aircraft body axes were used ($x+$ points out of the nose, $y+$ points out of the right wing, and $z+$ points down through the fuselage). The folding wings would fold along hinge lines parallel to the body x -axis and rotating wings would rotate about the body z -axis. A hybrid folding and rotating wing would fold sections of the wing about lines parallel to the x -axis and then rotate the wing about the z -axis. After running several MATLAB simulations, a STO-wing configuration was chosen. STO-wings allow nearly infinite wing length, as the wings can fold into the negative x -direction which has no geometric limit. The folding, rotating, and hybrid wing designs not only have shorter maximum spans but also have more

mechanical and electrical complexity. By using the STO-wing configuration, the largest constraint is shifted from the size of the wing box to the size of the nose-to-aft-gear limit. While the length of the wing in principle could be nearly infinite, increasing the wingspan beyond 2.5 m would shift the center of gravity (CG) of the aircraft beyond the two-foot nose to aft gear limit. As a result, this became the primary limiting factor. This constraint along with manufacturability concerns span limited the design to 2.5 m.

Considering the large range of operating velocities, the next design point was determining the optimal way to achieve a quick takeoff. High lift devices such as flaps and slats were considered baseline starting points. Flaps and slats would increase the operating C_L at takeoff without increasing the operating C_L during flight, allowing for more lift generation at slow speeds without the drag penalty at high speeds. Looking to actual STOL aircraft for inspiration, thrust vectoring motors and tilt wings were considered. The F-35B (Fig. 3.2.1) was a case study as it employs complex thrust vectoring to take off a ramp on the HMS Queen Elizabeth. Additionally, a tilt-wing case was considered where the wing would tilt upwards of 30 degrees with motors mounted on the wings. The Georgia Tech and NASA collaboration, “Greased Lightning”, had these capabilities and was studied as an additional way to operate over a large range of possible velocities [1]. Both aircraft mentioned above exhibited what WUDBF called “enhanced geometries,” which decreased the takeoff distance but did not affect cruise operations.



Fig. 3.2.1-2, 1. F-35B ramp takeoff in STOL mode [2]; 2. Greased Lightning in a hover [3].

3.5 Initial Aircraft Designs

After preliminary analysis, the design philosophy discussed above yielded seven aircraft designs that could meet the mission requirements and excel at competition. Each aircraft was designed in XFLR5 and examined a different geometry, propulsion, avionics, and store configuration. All seven aircraft were longitudinally stable, had a wingspan of 2.5 m, and had an H, W, or V tail to stay within the two-foot height limit. They also used NACA 6412 airfoils for its high $C_{L,0}$ and well defined aerodynamic characteristics. Each aircraft was assigned a risk ranking from zero (least risk) to six (most risk) based on aircraft stability during all portions of flight and manufacturability. The risk factors were decided via discussion between all four subteams and encapsulated the chance that the plane would be able to perform as expected, without failure, and by the competition deadline.

3.5.1 The *Anaconda*

The WUDBF *Anaconda*, modeled after the off-the-shelf *Ready Made RC Anaconda*, represents the archetypal range platform and uses a conventional straight tapered wing with leading edge slats and trailing edge flaps (See Fig. 3.4.2) [4]. A pusher propulsion setup with an inverted V-tail provided thrust and stability. The WUDBF *Anaconda* is slightly larger than its *Ready Made RC* baseline with a 2.5 m vs. 2.1 m wingspan and as a result, has proportionally larger tail surfaces and boom lengths [4]. WUDBF validated the simulator's results using a *Ready Made RC Anaconda* (Section 8.4.1). By validating the WUDBF *Anaconda*'s results with real-world data, such as takeoff distance, WUDBF was confident kinematic simulator predictions were accurate (See Section 4.3.2). The WUDBF *Anaconda* benefitted from a simple main wing and propulsion design, improving manufacturability. Since the *Anaconda* was already flight proven, the WUDBF *Anaconda* was viewed as the lowest risk aircraft with a risk score of 0. However, the WUDBF *Anaconda* did not include any enhanced geometries and therefore would not carry as many attack stores when compared to other models.

3.5.2 The *Bush Plane*

The *Bush Plane* (See Fig. 3.4.3) was modeled after the RC *Carbon Cub* [5] and the *Draco Bush Plane* [6]. This model attempts to emulate the short takeoff and landing capabilities as well as the high maneuverability of these aircraft. The *Bush Plane* used a single engine puller configuration with a straight tapered wing and H-tail (to stay within the height limit). For high lift devices, the *Bush Plane* used trailing edge flaps. The *Bush Plane* was the most traditional of the aircraft examined, and therefore its possible risk in terms of aircraft stability and manufacturability were low. A risk factor of 0.5 was applied. Similar to the *Anaconda*, the *Bush Plane* employed no enhanced geometries which limited its total store capacity.

3.5.3 The *P38*

The *P38* (Fig. 3.4.7) was modeled after the Lockheed Martin *P-38* and consisted of a twin-boom, twin-puller aircraft with an H-tail, a straight wing, and trailing edge flaps [7]. Additionally, the *P38* thrust vectored its motors upwards to lower its required takeoff speed (an enhanced geometry). Thrust vectoring the motors would present a unique challenge to aircraft stability as the thrust line of the motor must travel through the center of gravity of the aircraft to avoid a negative or positive pitching moment. Therefore, the motor hinge line must be mounted on the center of gravity line along the y-axis. Circular cutouts on the main wing would be required, as the propeller wash would simply hit the top of the wing and deflect. This would negate the advantage of the thrust vector motor. With the addition of thrust vectoring, the performance of the *P38* is higher than the *Anaconda* and the *Bush Plane*. However, the thrust vectoring of two motors and the required wing cutouts significantly increases both stability and manufacturing concerns. As a result, the *P38* was assigned a risk score of four.

3.5.4 The *Three Surface*

The *Three Surface* (Fig 3.4.5) is a dual-engine, straight-wing with an H-tail and additional canard. Similar to the *P38*, the engines on the *Three Surface* are thrust vectored, increasing aircraft performance.

However, by adding the canard, the neutral point of the aircraft is pulled forward. To maintain the same static stability, the center of gravity must shift forward. Since the center of gravity is shifted forward, the motor mounts must follow suit. With clearance from the wing, the motors can be thrust vectored without the need for cutouts. This reduces the mechanical complexity of the model but leads to other constraints. Specifically, with the center of gravity pulled forward and given the two-foot nose to aft landing gear constraint, it was nearly impossible to achieve the required center of gravity. Too much mass was too far aft. To solve this problem, the propulsion model size was decreased along with other subsystems. These actions, while necessary to maintain stability, decreased aircraft performance offsetting the thrust vectoring system. The small margin of error for stable flight coupled with the thrust vectoring of two motors resulted in a risk factor of 3 for the *Three Surface*.

3.5.5 The Spirit

The *Spirit* (Fig. 3.4.1) is a single-engine pusher whose enhanced geometry was modeled after the *F-35B*'s short takeoff capabilities [2]. Building off the successful (and validated) *Anaconda* model, the *Spirit* has a tapered swept wing with an H-tail as well as trailing edge flaps. By sweeping the wing back, the neutral point and the center of gravity are moved further back to obtain the same static stability as a traditional aircraft. This allows the aft motor to rotate about the center of gravity, enabling thrust vectoring. Since the motor is mounted behind the body, it rotates down, and as a result, there is no wing propeller wash interference. Additionally, the motor can be mounted directly to a flap on the back of the fuselage using a single servo, decreasing complexity compared to the minimum two servos needed to rotate each motor on earlier aircraft. The thrust vectoring significantly increases the takeoff performance of the pusher. The *Spirit* receives a low risk factor of 2 since it is essentially a modified *Anaconda* with an increased complexity due to the thrust vectoring.

3.5.6 The Pusher Puller

The *Pusher Puller* (Fig. 3.4.6) is a dual engine with a front and back mounted motor, a tapered wing, and an H-tail. Expanding on the *Anaconda* design, the *Pusher Puller* uses thrust vectoring on both motors to accomplish short takeoffs. With each motor causing a moment in the opposite the direction, the thrust lines of the motors did not have to fall through the center of gravity. However, their moments had to counteract each other throughout the takeoff up until normal flight is achieved. While this plan solves the moment problem — and allowed for more traditional placements of the motors and center of gravity — the complexity of ensuring the moments caused by the aft motor would cancel the moments caused by the front motor were immense. The *Pusher Puller*'s high performance come with a high risk score of 5.

3.5.7 The Greased Lightning

The final aircraft studied was the *Greased Lightning* which was modeled after a joint study between Georgia Institute of Technology (GIT) and NASA Langley Research Center. The *Greased Lightning* used rotating wings to change the thrust angle of several motors mounted on both the main wing and horizontal stabilizer [1]. This allowed for vertical takeoffs and landings (VTOL). While VTOL was

banned under DBF competition rules, the original *Greased Lightning* design could be modified for rolling takeoffs [8]. In WUDBF's design, the *Greased Lightning* rotated its wing upwards to 30° , providing thrust vectoring that would also turn the entire wing into a flap. This design had extremely high performance due to its high lift and thrust vectoring, but its technical complexity was immense. One device — as a worm gear — would have to rotate the entire wing, wing locking mechanism, and nearly all of the attack stores up and down 30° . Additionally, rotating the wing up to 30° would necessitate leading edge slats to prevent a stall, and landing would necessitate flaps. This aircraft received the highest risk score of 6.

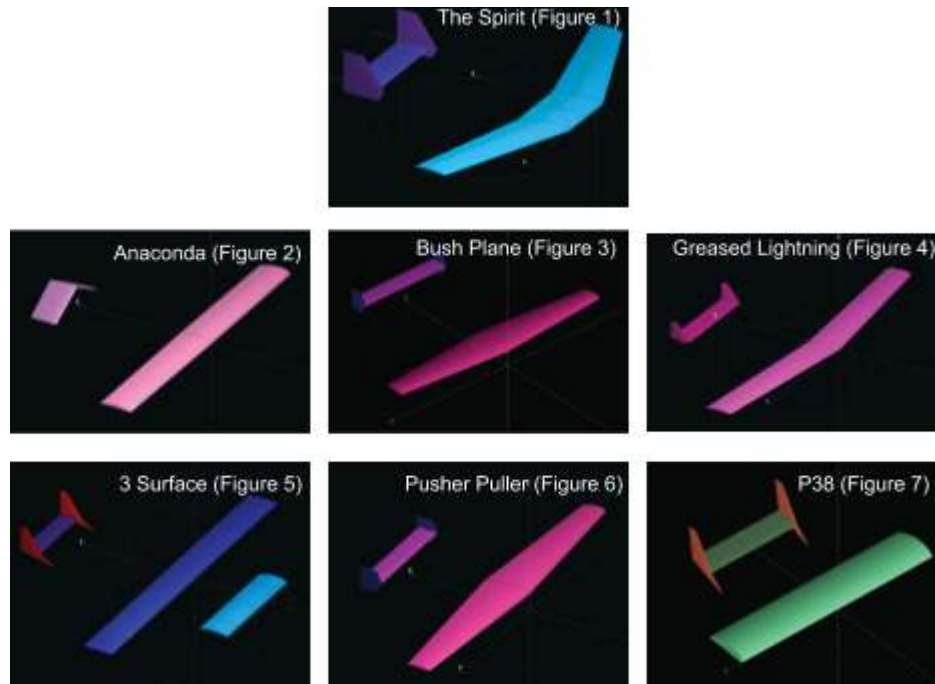


Fig. 3.4.1-7 Initial conceptual designs for various aircraft.

3.6 Final Design Selection

KM Sim (a mission model Matlab simulation), documented in Section 4.2.2, was used in the conceptual design to easily compare models. After running the simulator and iterating through mass configurations to optimize takeoff weights, the final aircraft design selection was made. Figure 3.5 displays the Mission 3 score for all aircraft compared to each aircraft's assigned risk. Since Mission 3 requirements were higher than the mission requirements for Missions 1 and 2, the highest aircraft performance on Mission 3 would translate to high aircraft performance on Missions 1 and 2. Three aircraft immediately stood above the rest. The *Greased Lightning* earned a score of 16 points, meaning it dropped 16 stores flying 16 laps during Mission 3. Additionally, the *Spirit* and the *Pusher Puller* earned an initial score of 15 points, dropping 15 stores and completing 15 laps. The *Pusher Puller* was immediately discarded from the design options, as it had a similar potential score as the *Spirit* with a far higher risk (2 vs. 5). Simply looking at score would result in the *Greased Lightning* as the final selected design, but the

risk was significantly higher. The *Spirit* was chosen as WUDBF's flight competition vehicle, as its risk factor was significantly lower than aircraft performing at similar levels.

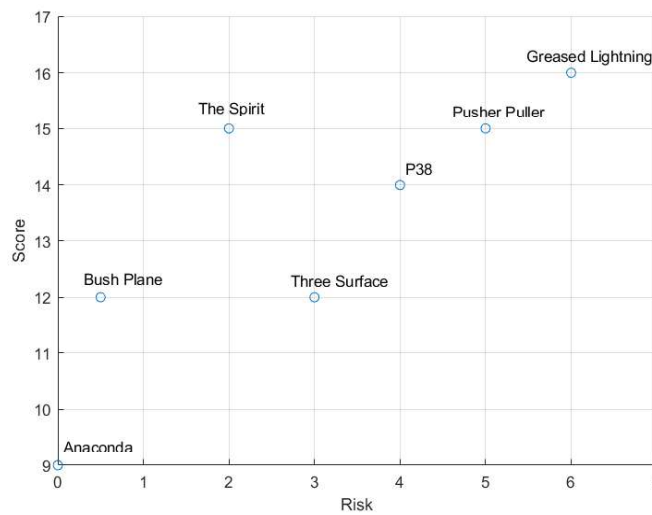


Fig. 3.5 Scoring plot of Mission 3 score vs. aircraft risk.

4. Preliminary Design

4.1 Aerodynamic Design and Analysis

After the *Spirit* was chosen as the primary flight configuration, the design was iteratively improved to ensure the plane was stable, controllable, and achieved maximum performance. A combination of XFLR5, OpenVSP, and Matlab was used in aircraft analysis. The size and shape of the main wing and stabilizers were driven by iterative static and dynamic stability analyses. Control surfaces and high lift devices were installed to ensure controllability and minimize takeoff distance respectively.

4.2 Modeling and Simulation

4.2.1 Drag Determination

While XFLR5 provided accurate lift and induced drag measurements, the analysis considered only inviscid flow. Parasitic drag, as well as drag caused by unmodeled items such as the fuselage, stores, and other components, were not included in the model. To factor in these additional components, a parasitic drag script was created to estimate the drag considering objects as thin plates. The thin-plate drag estimate was then corrected for thickness using wind tunnel data correlations. Additionally, ANSYS Fluent analysis was run on a store to calculate its drag coefficient, $8.041\text{E-}4$, using frontal reference area. (See Fig. 4.1). This data was fed into the Mission Model.

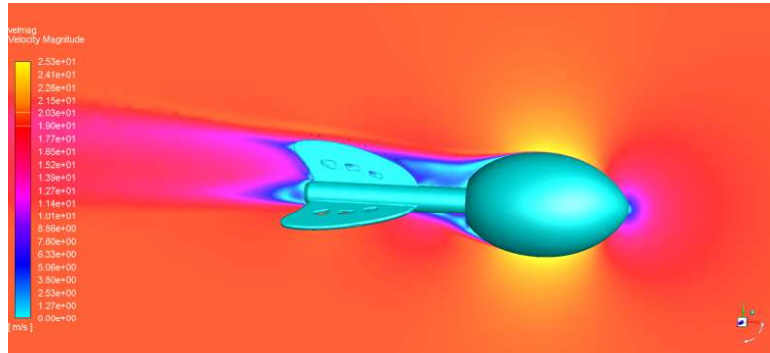


Fig. 4.1 Attack store drag model.

4.2.2 Mission Modeling

After the aircraft was designed in XFLR5, the aircraft aerodynamic properties, inertial properties, and loadout were programmed into a struct used by Kinematic Mission Simulator (KM Sim). KM Sim is an idealized control dynamics simulation of the aircraft using a prescribed course programmed in MATLAB. The simulator is used to assess aircraft kinematic performance in a faster-than-real-time simulation.

Simulated flight phases include, but are not limited to, ground roll, rotation, optimum climb, max climb, cruise, constant radius and constant g-load turns, and optimum glide. Each flight phase is modeled as a separate series of differential equations, defining 14 states based on desired aircraft behavior.

KM Sim can execute two natively defined post-processing scripts and any user-specified post-processing script for post-simulation analysis and scoring. The first native post-processing script, a trajectory plot, shown in Fig. 4.2, displays the complete flight path of an aircraft.

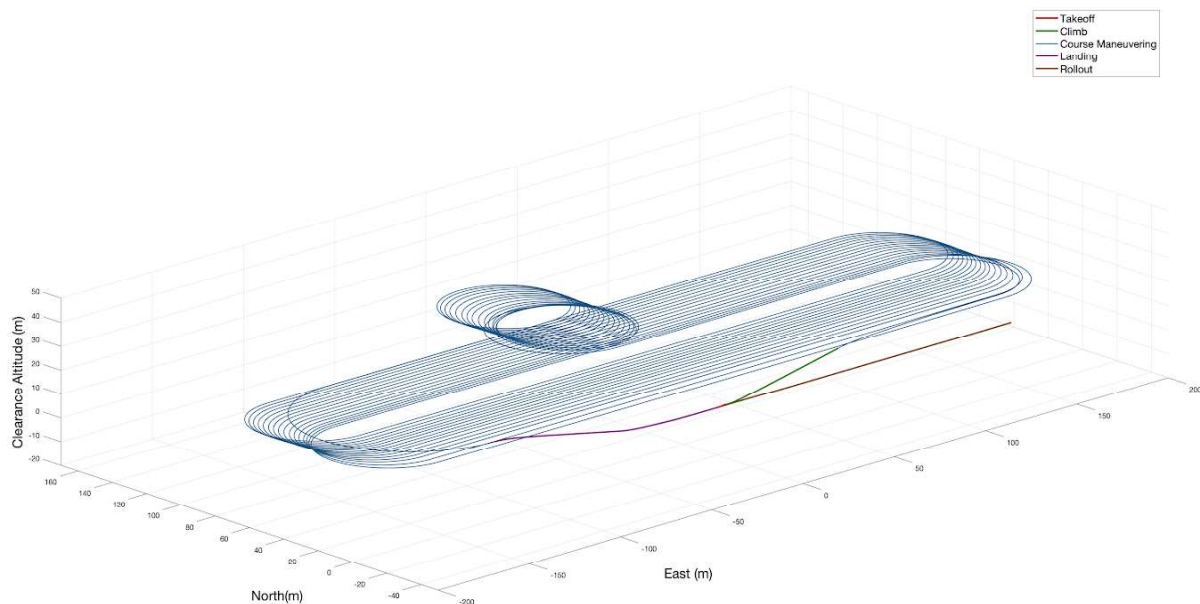


Fig. 4.2 Trajectory plot of KM Sim results.

Trajectory plots are color-coded by takeoff (red), climb (green), course maneuvering (blue), landing (magenta), and rollout (brown). They are plotted within an East-North-Up coordinate space.

The second native output, shown in Fig. 4.3, is an eight-panel results summary of the flight.

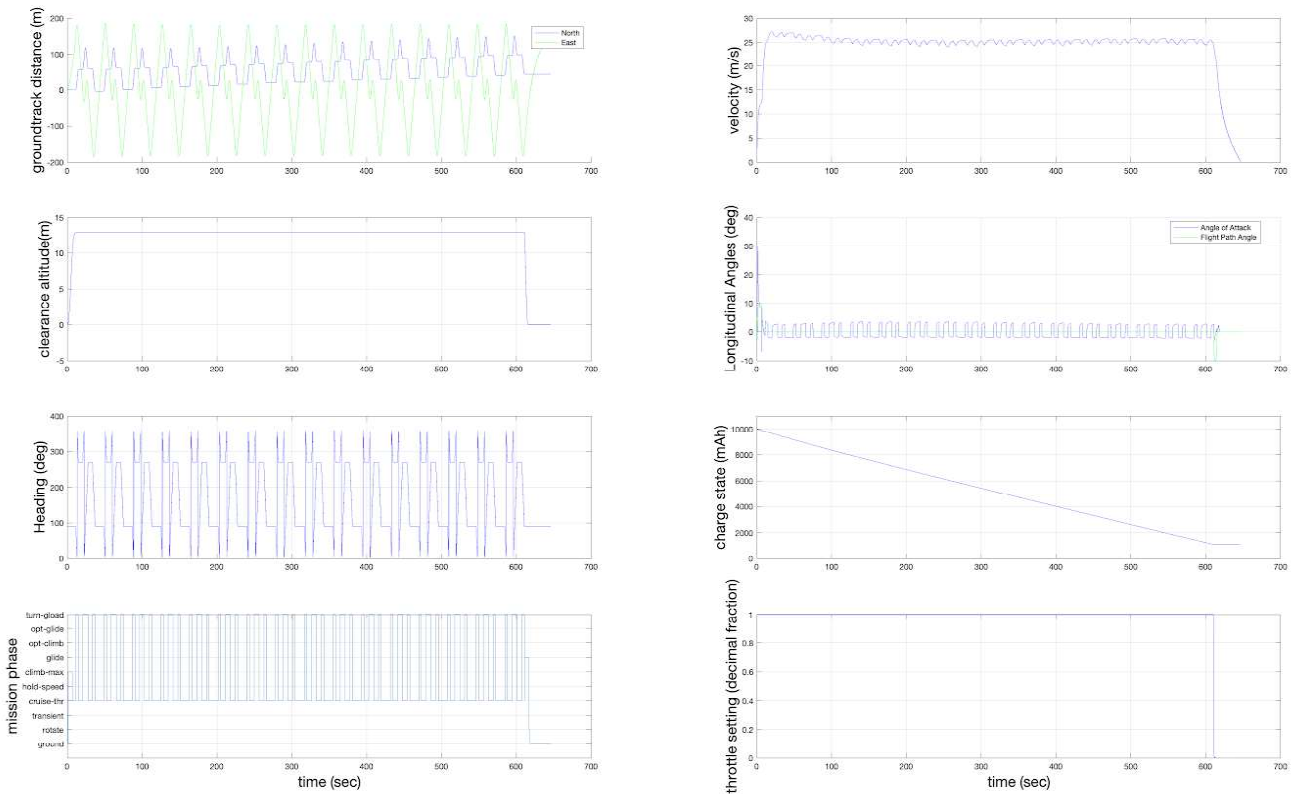


Fig. 4.3 Example of eight panel results summary.

This summary contains time history plots detailing flight information such as aircraft position, heading, the angle of attack (AoA), velocity, and altitude from the dynamics modeling and throttle and battery charge state from the integrated propulsion model.

KM Sim has the necessary features to simulate each of this year's missions. For example, KM Sim can simulate takeoff from a ramp and can detect post-ramp runway strikes. It also supports aircraft configuration changes, such as thrust vectoring and flaps, along with loadout changes, such as store drops. The aircraft aerodynamics and propulsion models are table lookup values within the plane struct. Configuration changes are performed by switching the aerodynamic table lookup. Loadout changes, on the other hand, are simulated by changing the plane weight and drag within the plane struct.

KM Sim can be employed for a wide variety of studies. For instance, aircraft designs can be run through different courses with the same initial state and through the same course with a different store, weight, flap, and thrust vectoring instructions. Additionally, parameter studies, range calculations, and maximum takeoff weight calculations are all easily performed in KM Sim. A risk assessment study performed in KM Sim can be found in Section 3.6.

4.3 Main Wing Planform Area

To size the planform area of the main wing, two variables were considered: the cubic wing loading and generating enough lift at cruise speed and cruise AoA to support the vehicle's AUW. Given that the sensitivity study indicated a final score proportional to wingspan, the wingspan was set at 2.5 m. This is the longest span WUDBF had the resources to manufacture. From an XFLR aerodynamic analysis, the cruise C_L with an Eppler 393 airfoil (see Section 5.2 for airfoil selection) was 0.4 at a zero degree AoA. From KM Sim it was determined the desired cruise speed was 25 m/s since that allowed for the completion of over 16 laps even if the plane was only carrying 15 stores. This gives the team some room for pilot contingencies. Although the estimated AUW of the aircraft is 80 N, a safety factor was included so the design had an AUW of 115 N. This additional margin accounts to potential lift spoilage due to structural components and the attack stores.

Cubic wing loading (CWL) is a parameter often calculated by model aircraft pilots to determine how "flyable" model is. Unlike wing loading, its implications apply to different size aircraft. This is to say that two aircraft with vastly different wingspans but equal CWL factors will handle similarly, while two aircraft of vastly different wingspans but equal wing loadings will most likely fly dramatically differently. Given the capabilities of the team's pilot, WUDBF is comfortable selecting a CWL factor of 14.3-14.67 which is representative of sport aircraft [9].

Given the above considerations, a wing area of 0.872 m² was selected, which yields 117 N of lift at cruise, a CWL factor of 14.4, and an aspect ratio (AR) of 7.18.

4.4 Main Wing Sweep, Longitudinal Stability, and Horizontal Stabilizer Sizing

Longitudinal stability is fundamental to aircraft design and this year's AIAA competition. Since the *Spirit* is dropping an attack store each lap, the aircraft must be longitudinally stable with a shifting center of gravity. Favorable longitudinal handling is also crucial in preventing dangerous longitudinal oscillations during takeoff which is the most dangerous flight condition due to low control power. Longitudinal stability is dependent on several factors which are shown in Eq. 4.1 [10].

$$C_{m_\alpha} \equiv \frac{\partial}{\partial \alpha} C_m = C_{L_{\alpha w}} \frac{l_{acw}}{\bar{c}} + C_{L_{\alpha h}} \left(1 + \frac{\partial \varepsilon}{\partial \alpha} \right) \eta_h \frac{S_h}{S} \frac{l_{ach}}{\bar{c}} \quad (4.1)$$

As shown above, the distance from the aerodynamic centers of the main wing and horizontal stabilizer ($l_{ac,w}$, $l_{ac,h}$) to the center of gravity, as well as the size of the horizontal stabilizer S_h are variables that affect longitudinal stability. The center of gravity location was fixed on the hinge line of the thrust vectoring mount to avoid possible pitching moment. Due to the two-foot height limit and to avoid a propeller ground strike, the hinge line for the thrust vectoring flaps was 0.281 m away from the leading edge of the wing. As previously mentioned, sweeping the main wing aft would increase the magnitude of $l_{ac,w}$ (making the value more negative) increasing the stability of the aircraft. A Microsoft Excel function approximated the distance of the aerodynamic center from the leading edge of the wing. This, coupled

with XFLR5 iterations, found sweeping the wing back at 20.4° defined from the quarter chord (putting the aerodynamic center 0.436 m from the leading edge of the wing) in combination with elevator size and placement resulted in a $C_{m,\alpha}$ of -1.39 (see Fig. 4.4). Elevator distance was driven by longitudinal eigenvalue analysis seen in Section 4.6. Based on prior aircraft designs of similar size and team experience, $C_{m,\alpha}$ between -0.5 and -1.8 were appropriate. The -1.39 $C_{m,\alpha}$ fit within the bounds of the guidelines; this was key since the plane would have a variable center of gravity during flight as stores were dropped and therefore the $C_{m,\alpha}$ would shift during flight. To ensure stability in all flight regimes, longitudinal stability analysis with different numbers of stores dropped was completed in Section 4.4.1.

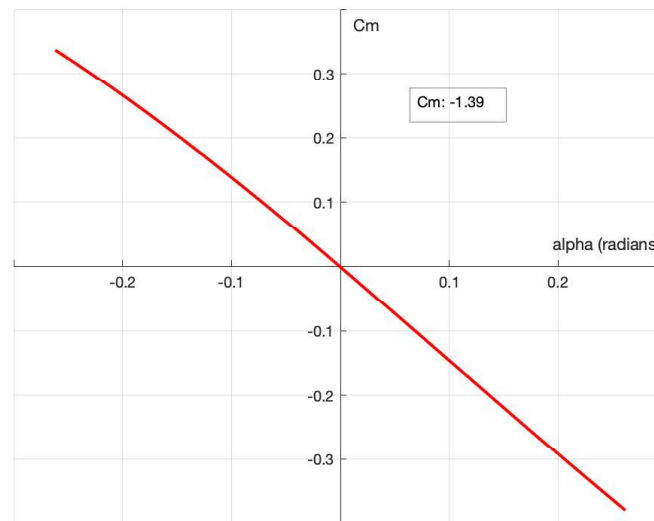


Fig. 4.4 Longitudinal stability vs. alpha.

4.4.1 Store Placement and Longitudinal Stability with Various Sets of Stores Dropped

The attack stores were placed to minimize stability issues and to minimize drag. To minimize drag, they were placed in front of or behind landing gear, which already causes drag. The stores could also not be placed in front of the propeller, since the store could collide the propeller.

To ensure the *Spirit* was stable in all conditions, the team tested stability with various configurations of attack stores attached. The C_m vs. α graphs are recommended to have a slope between -0.5 and -1.8 for this aircraft [6]. To get the data, stores were removed from front to fore, which would result in the steepest slope, and then from aft to fore, which would result in the lowest slope. This test determined if the plane would be stable under extreme circumstances where stores were dropped in a detrimental order; these conditions are not what the team plans to replicate in the competition. Tables 4.1.1 and 4.1.2 show the slopes of the C_m vs. α graph depending on which masses were removed, and Fig. 4.5 displays the locations of the numbered stores. In the case of two stores stacked vertically, the top number represents the top store.

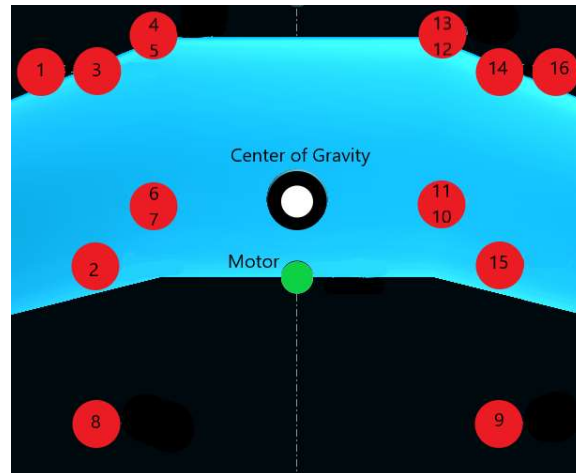


Fig. 4.5 Numbered store locations on the underside of the *Spirit* in the x-y plane.

Tables 4.1.1-2 Slope of C_m vs. α for different masses removed from fore and aft.

1. Fore		2. Aft	
Stores Removed	Slope [rad^{-1}]	Stores Removed	Slope [rad^{-1}]
No Stores	-1.46	8, 9	-1.60
4, 5, 12, 13	-1.01	8, 9, 2, 15	-1.65
4, 5, 12, 13, 1, 3, 14, 16	-1.07	8, 9, 2, 15, 6, 7, 10, 11	-1.64
4, 5, 12, 13, 1, 3, 14, 16, 6, 7, 11, 10	-0.968		

4.5 Directional and Lateral Static Stability

Directional and lateral stability modes were determined after completing longitudinal stability analysis. Since the lateral directional stability space links lateral and directional modes, individually looking at their static stability derivatives does not provide a complete view of the aircraft dynamics. An analysis of eigenvalues, undamped, and damped frequencies completed in section 4.7 was deemed more useful for understanding directional and lateral stability. For reference, the lateral and directional static stability is as follows. Roll stability is slightly negative with a C_{l_b} of -0.0568 rad^{-1} due to sweep in the main wing. The directional stability, C_{n_b} , is 0.209.

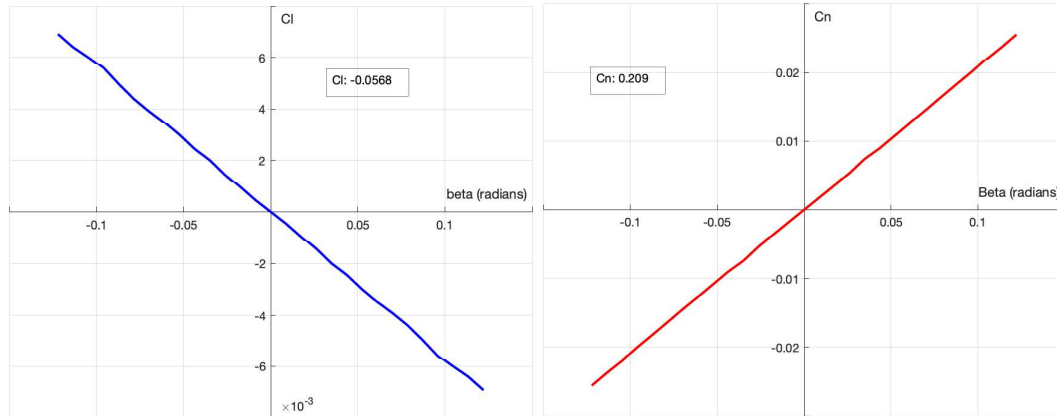


Fig. 4.6.1-2, 1. (Left) Lateral stability vs. (left) beta; 2. (Right) Directional stability vs. beta (right) respectively.

4.6 Longitudinal Eigenvalue Analysis

Longitudinal eigenvalue analyses conducted in XFLR5 shown in Fig. 4.7 drove horizontal stabilizer distance from the center of gravity. When plotted on the real and imaginary axes, The *Spirit's* complex roots for longitudinal stability case lie in the left (negative) half plane, which is necessary to ensure stability [10]. WUDBF chose to constrain analysis to the short period mode as the long period mode has a high enough time scale such that minor pilot control inputs can eliminate any oscillations [10]. Iterating through a range of tail distances, WUDBF determined the proper tail distance by comparing the damped and undamped frequencies to those of other aircraft (see Fig. 4.8). The final shape of the longitudinal eigenvalues (Fig. 4.7) is similar to other aircraft and serves as additional confirmation that aircraft dynamics have been predicted correctly [11].

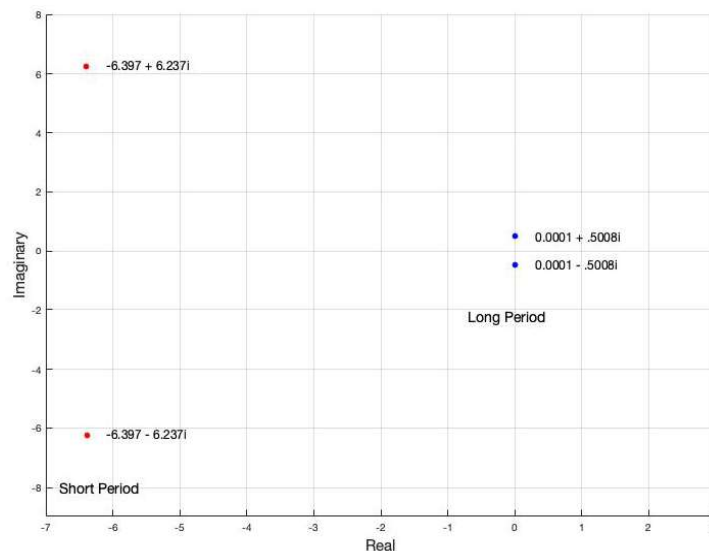


Fig. 4.7 Longitudinal eigenvalues characteristic short and long period modes for final horizontal stabilizer.

Table 4.2 displays the short period and long period natural and damped frequencies obtained from the eigenvalue analysis in Fig. 4.7.

Table 4.2 Short period and long period natural and damped frequencies.

Mode	Natural Frequency (Hz)	Damped Frequency (Hz)	Damping Ratio
Short Period	1.420	0.882	0.716
Long Period	0.080	0.080	0.0002

The *Spirit's* natural frequency and damping ratio of the short-period mode fall along the boundary between the satisfactory and unsatisfactory region for acceptable handling qualities (see Fig. 4.8). However, this thumbprint, pulled from [11], was created under guidelines for acceptable handling qualities for larger aircraft. Since the *Spirit* is a small model plane, pilots are accustomed to flying airplanes that respond sensitively to pitch changes. Therefore the longitudinal dynamics of the *Spirit* will still produce adequate handling qualities despite its higher short period natural frequency. This viewpoint is confirmed in Fig. 4.9 where the natural short period frequency is noticeably higher for aircraft of smaller span [11].

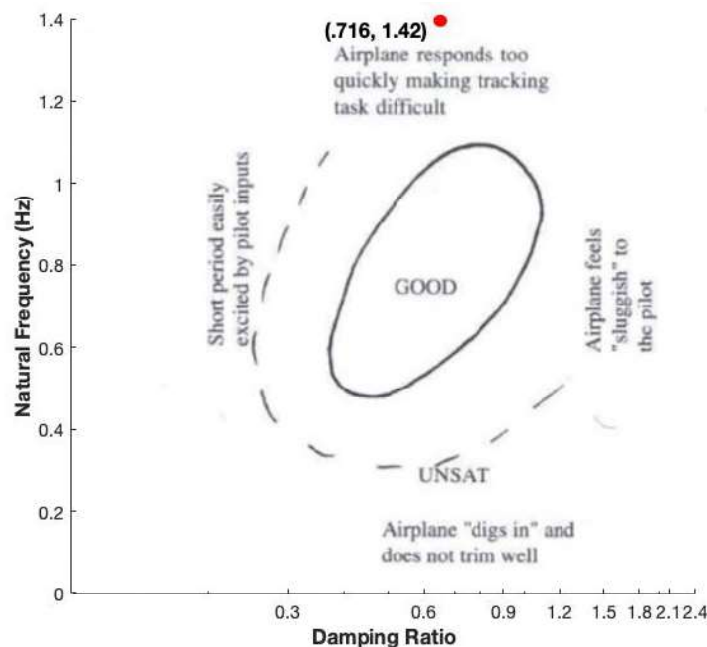


Fig. 4.8 Acceptable Natural Frequencies vs. Damping Ratios [11] with the *Spirit's* Ratio Overlaid (Red Dot).

The *Spirit's* short period mode natural frequency plotted versus wingspan follows the trend of a higher natural short period frequency for a smaller aircraft [11].

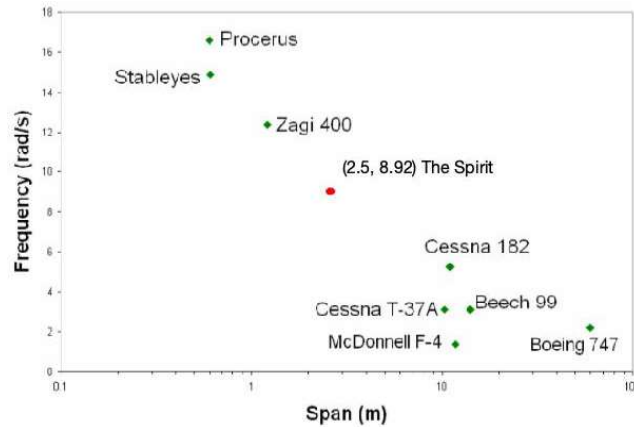


Fig. 4.9 Natural short period frequency vs. wingspan for various aircraft [11].

4.7 Lateral-Directional Eigenvalue Analysis

Vertical Stabilizer sizing was driven by the lateral-directional eigenvalue analysis. The Dutch roll stability was initially low compared to similar aircraft so an additional third tail was added to increase dampening [11]. The Dutch roll stability characteristics were then compared to other aircraft in Fig. 4.11 to ensure the *Spirit's* eigenvalues were in line with similar aircraft. As shown in Fig. 4.10 and Table 4.4, the spiral mode time-to-double and the roll mode time-to-half are extremely large and small respectively. Having a high time to double for spiral mode and low time to half are characteristics of a laterally stable aircraft so no design changes were made based on these two roots [10]. The *Spirit's* lateral-directional eigenvalues will produce dynamic stability in addition to satisfactory handling qualities.

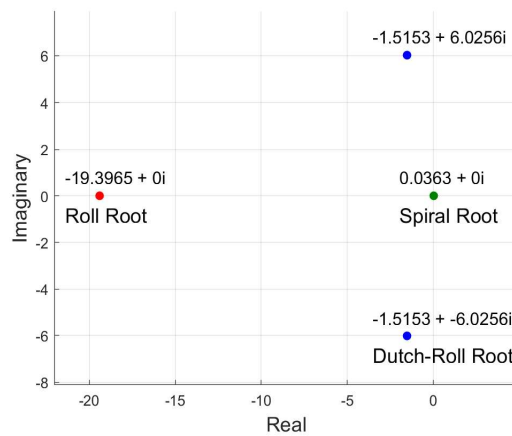


Fig. 4.10 Lateral directional eigenvalues: roll root, spiral root, and Dutch root.

Table 4.3 displays the characteristics of the Dutch roll eigenvalue.

Table 4.3 Dutch roll natural frequency, damped frequency, and damping ratio.

Mode	Natural Frequency (Hz)	Damped Frequency (Hz)	Damping Ratio
Dutch Roll	0.950	0.923	0.236

Table 4.4 displays the characteristics of spiral and roll eigenvalues.

Table 4.4 Spiral and Roll Time To Double and Time to Half Respectively

Mode	(s)
Spiral	Time To Double: 19.175
Roll	Time to Half: 0.0036

The *Spirit's* Dutch roll frequency graphed versus wingspan follows the trendline of other stable aircraft [11].

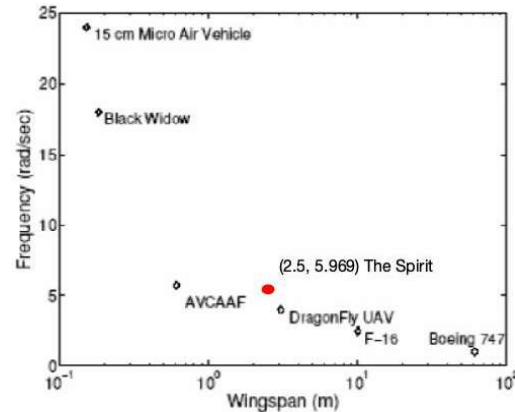


Fig. 4.11 Dutch roll natural frequency vs. wingspan size overlaid with other aircraft [11].

4.8 Wing Design for Aerodynamic Performance

Span limited aircraft when fitted with an elliptical lift distribution have the minimum induced drag; this distribution is common in most conventional aircraft [12, 13]. For the main wings, the team applied twist and tapering to approximate an elliptical lift distribution. Along the span, the twist was applied at three locations: 1° at 0 mm, 1.9° at 235 mm, and -2.5° at 1250 mm. The actual lift distribution closely models the ideal elliptical shape as shown in Fig. 4.12, and manufacturability was increased with only three spanwise twist locations. Additionally, by de-twisting the wing at the outer-spanwise locations the aircraft would have increased roll control throughout a stall and weaker tip-stall tendencies.

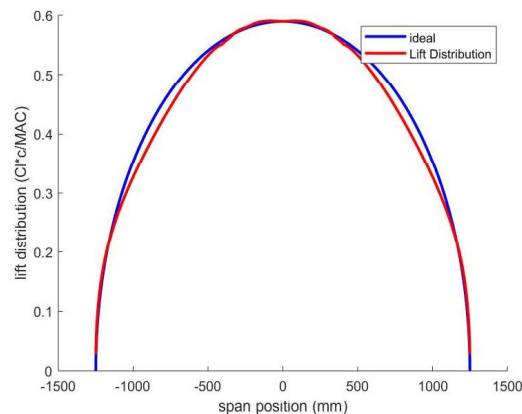


Fig. 4.12 Ideal vs. actual elliptical lift distributions of the main wing.

4.9 High Lift Devices

To generate enough lift on takeoff, WUDBF employs a series of high lift devices. Slotted flaps were implemented following guidelines from *Theory of Wing Sections* by Abbott and Von Doenhoff where the chord length was 0.30 of the total chord and the max deflection angle was 41° [14, pp. 207-208]. Although slotted flaps did not increase lift as much as Fowler flaps, WUDBF was limited by manufacturability (See Section 5.5.3). Additionally, the thrust vectoring motor was set on a flap similarly sized to the slotted flap representing 0.30 of the chord length. However, it did not use a slotted setup since there were additional constraints from the motor mount. Both the length of the thrust vectoring mount and the slotted flap were determined by rule of thumb and other aircraft of similar designs. To further increase lift, the ailerons on takeoff would be set as flaps. As a result of all these high lift devices, the max C_L is estimated to be 2.52. This estimate came from interpolating 2D airfoil lift and drag data from *Theory of Wing Sections* with reconciliation from NASA wind tunnel data on fowler flaps [14, 15].

4.10 Control Surface Sizing

Given the relatively traditional design of this aircraft, all the control surfaces (rudder, elevator, and ailerons) were designed using rule of thumb parameters that worked well on past aircraft designs of a similar nature. To confirm the sizing was correct, a Realflight 8 remote control plane model of the *Spirit* was created. The handling qualities in the Realflight 8 Model met WUDBF's requirements for roll rate, pitch rate, and yaw rate.

4.11 Initial Performance Estimates

After significant optimization and analysis, WUDBF was able to increase the performance of the *Spirit* so that on Mission 3 it is able to take off with 16 stores and fly 16 laps dropping all 16 stores. This is a single store increase over the initial projected store capacity predicted in the conceptual design phase. KM Sim projects the *Spirit* is able to take off by the end of the ramp at about 5.5 m/s. Mission 1 is projected to take about 2.5 minutes including takeoff where the plane will fly a lap in about 45 seconds. Mission 2 is projected to take about 100 seconds with each lap taking about 32 seconds. In Mission 2 the *Spirit* has been designed to withstand a 5g turn structurally (see Section 5.3.1) and without stalling.

Fig. 4.13 is the preliminary design of the *Spirit* in OpenVSP after performing the analysis from above. Note the radome has not been described as that is a detailed design point in Section 5.5.9.

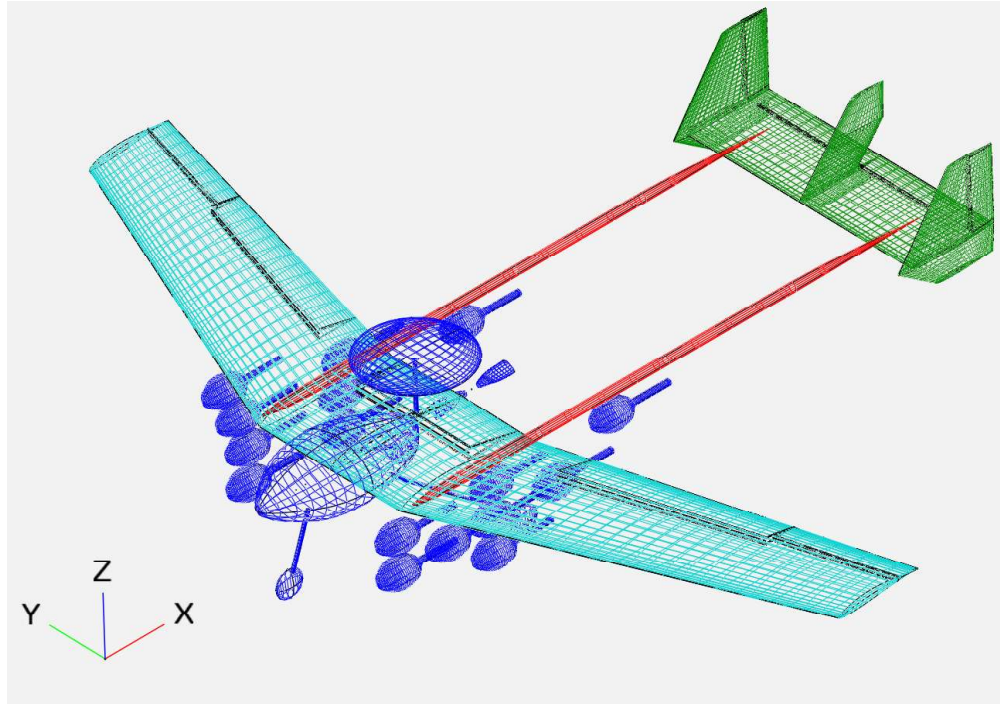


Fig. 4.13 Preliminary design of the *Spirit*.

4.6 Propulsion Sizing

The propulsion system on the *Spirit* maximizes cruising speed and still allows the aircraft to complete all three flight missions. Higher cruising speed increases the number of payloads that can be dropped in the allotted time, which is the most important factor in the score (see Section 3.3). Cruising speed is directly dependent on dynamic thrust, so the goal of the propulsion system is to maximize thrust at cruising speed or dynamic thrust, with two key constraints:

1. Static T/W: Ensure static T/W is high enough to take off with all payloads in the allotted 10 ft ramp.
2. Endurance: Ensure cruising efficiency is high enough for the propulsion system to last the 10-minute time window on Mission 3 (the longest mission).

4.6.1 Overall Sizing

The minimum static T/W to avoid collision with the ground after takeoff was found to be approximately 0.65 via KM Sim. Because the maximum AUW of the plane was determined to be 8kgf from aerodynamic and structural constraints, the minimum thrust was determined to be:

$$\text{Min Static Thrust} = AUW_{max} \cdot (T/W)_{min} = 5.2 \text{ kgf} \quad (4.1)$$

with the addition of a small factor of safety results in a desired thrust of 6 kgf. The cruising efficiency constraint was then applied. The total mechanical energy required for the propulsion system is the product of dynamic thrust, speed, and flight duration, all divided by mechanical efficiency. This relation implies that as stored energy increases, dynamic thrust and speed will increase.

The amount of electrical energy stored is limited by battery mass, which is in turn constrained by structural mass and AUW. A fuel-mass-fraction of 40%, similar to long haul airliners such as the Airbus A350 [16] and Boeing 777 [17], was chosen as a reasonable first-order estimate for the battery mass; this fraction was later confirmed by CAD structural mass calculations. An upper limit on battery weight was determined to be around 3.2 kgf.

The specific energy, or energy per unit mass, of quality NiMH battery cells ranges from 75 to 90 Wh/kg (see Fig. 5.5). Total stored electrical energy is the product of specific energy and battery mass, so feasible total electrical energy ranged from around 240 to 290 Wh.

4.6.2 Propeller Sizing

The propeller was chosen to optimize efficiency, maximize dynamic thrust, and prevent large efficiency decreases due to blade stall upon takeoff. Disk actuator theory, which provides basic propulsive relationships, holds that [18, pp. 346]:

$$\text{Mechanical Power} = \sqrt{\frac{2T^3}{\pi\rho d^2}} \quad (4.2)$$

where T is thrust [N], ρ is air density [kg/m³] and d is propeller diameter [m].

Power requirements decrease for a given thrust, with increasing propeller diameter. Mechanical efficiency is maximized with a large propeller. Therefore, the propeller on the *Spirit* is as large as it can be within geometric constraints. Conversely, mechanical efficiency is maximized with the minimum number of propeller blades, so a 2-blade propeller was chosen.

The propeller pitch was chosen to maximize dynamic thrust within reasonable static power requirements. Dynamic thrust is approximately inversely proportional to speed:

$$\text{Dynamic Thrust} \approx \text{Static Thrust} \cdot \left(1 - \frac{\text{Speed}}{\text{Pitch Speed}}\right) \quad (4.3)$$

As speed increases, dynamic thrust decreases to 0, at which point the pitch speed has been reached. Therefore, an increase to the pitch speed increases dynamic thrust at a given velocity, allowing the plane to fly faster. Since pitch speed scales linearly with propeller pitch, it would seem that propeller pitch should be maximally increased. However, large pitch-diameter ratios cause propeller blades to stall at low airspeed, requiring more power to get the same thrust (Fig. 4.14).

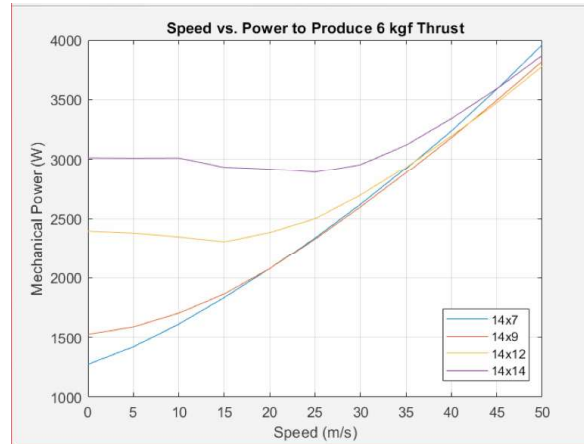


Fig. 4.14 Plot of mechanical power over airspeed required to produce 6 kgf of thrust. Made using QProp, a propeller analysis tool.

The propeller pitch ratio then must maximize dynamic thrust while still allowing for takeoff. Decreasing pitch ratio showed diminishing returns around 0.5, so it was not decreased any further, giving a propeller with a 15 in diameter and 8 in pitch.

4.6.3 Motor Sizing

The motor was chosen to maximize thrust given the energy constraints. The characteristics of motors are typically defined by their voltage constant, K_V :

$$K_V = \frac{RPM}{V}, \quad (4.4)$$

where RPM is rotational velocity in rotations per minute and V is the supplied peak voltage.

A custom first order motor simulation based on a model described by Mark Drela [19] was developed to iterate over motor K_V and other parameters.

Holding static thrust constant, the static and dynamic thrust per unit power, or electrical efficiency, peaks at some K_V , as shown in Fig. 4.14.1. This maximum electrical efficiency, however, occurs at a very low K_V . To achieve the desired static thrust of 6kgf at such a low K_V , a very high DC voltage is required. This voltage requires the battery pack to have a large number of NiMH battery cells in series, as seen in Fig. 4.14.2. However, most off-the-shelf ESCs and other RC plane electrical components only go up to the equivalent voltage of an 8S LiPo battery: a 23S NiMH battery. Also, larger numbers of NiMH cells in series can make charging the battery pack more difficult because chargers have a voltage limit.

With a 23S NiMH battery, the motor that generates the most thrust per unit power is between 400 and 450 K_V .

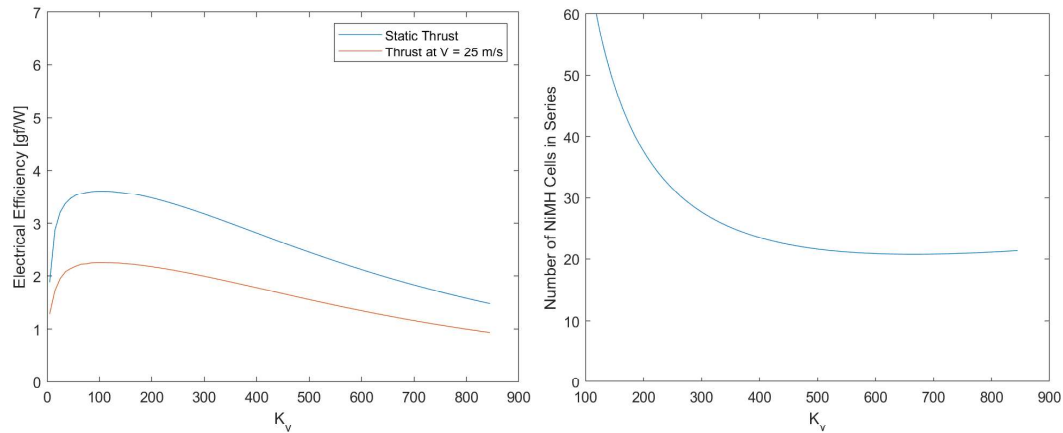


Fig. 4.14.1-2 1. Plot of electrical efficiency versus motor voltage constant for a static thrust of 6 kgf; 2. Number of NiMH battery cells in series required to achieve 6kgf of static thrust for different motor voltage constants.

4.6.4 Battery Sizing

Although the 23S battery pack was considered, using an even number of cells in series increased the flexibility of the battery pack design, so a 22S pack was chosen. Since the battery pack energy requirement was set at 240-290 Wh and the nominal voltage of the 22S pack is 26.4 V, a charge capacity of 9.1 Ah to 11 Ah was required.

Energy and power requirements of the motor were determined to verify this requirement. Electrical energy was found from the power predicted by the first order motor model and charge capacity for the 22S pack found for the 6 kgf static thrust, shown in Fig. 4.15. Static thrust requirements were temporarily relaxed to meet the charge capacity constraints and a 300 K_v motor was preliminarily chosen.

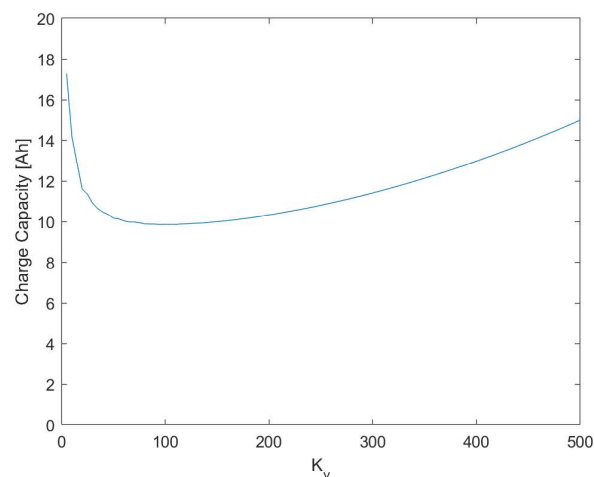


Fig. 4.15 Plot of required electrical energy over K_v for a 6 kgf static thrust.

Certain power requirements must be met for steady flight. Power was calculated to be 2000 W by running the first order motor simulation with the determined propeller and motor.

The battery cell analysis in Section 5.5.6 found the most practical battery pack to meet these power and energy soft requirements was a 2P pack of 5 Ah cells. The battery pack was preliminarily chosen to be a 22S2P pack, with a total charge capacity of 10 Ah and weight of 3.08 kgf.

4.6.5 Propulsion Verification

A time domain propulsion simulation found that 6 kgf of static thrust could be produced with lower than expected electrical power requirements.

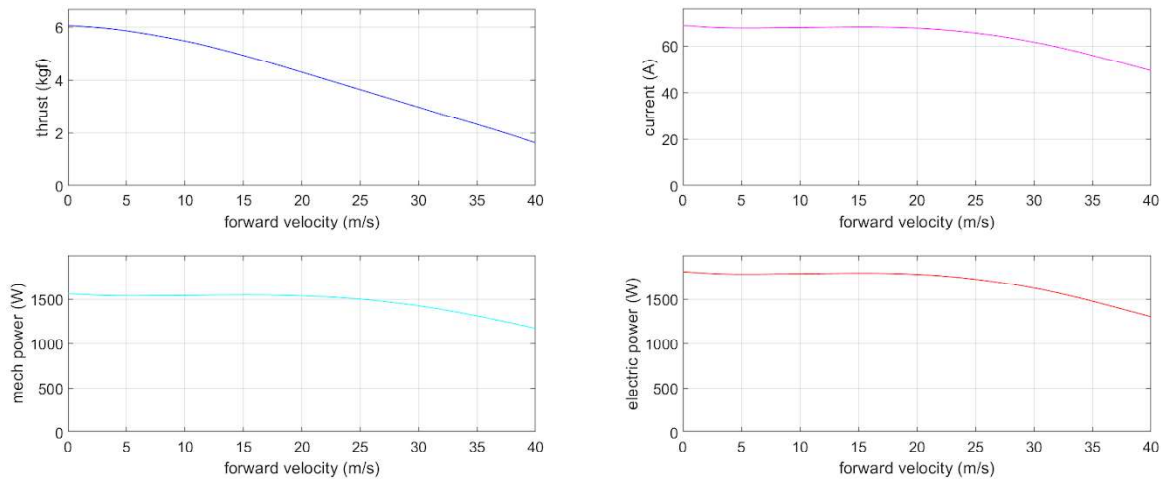


Fig. 4.16 Propulsion simulation results from time-domain motor simulation.

The motor K_v was iterated upon with the propulsion simulation and the optimal K_v was found to be 380 RPM/V. This propulsion package, the results of which are shown in Fig. 4.16, still leaves 1.1 Ah of charge capacity left in the battery, shown in Fig. 4.17.

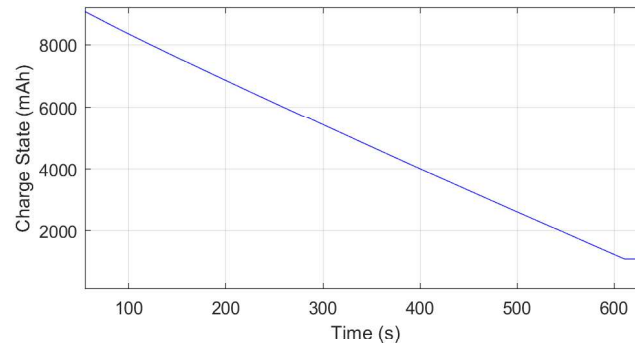


Fig. 4.17 Battery charge state over time for Mission 3 in the KM Sim.

The finalized propulsion package is shown in Table 4.5:

Table 4.5 Propulsion package.

Motor	Battery Pack	Propeller
Sunnysky X4130 380K _v	22S2P NiMH Turnigy 5000 mAh	APC 15x8E

5. Detail Design

5.1 Dimensional Parameters

Table 5.1 contains detailed information about the wing and control surface geometries, while Table 5.2 provides data regarding the weight, take off distance, and other hindrances on the *Spirit's* flight capabilities.

Table 5.1 Wing and control surface geometry.

Wing Geometry Data	
Wingspan (m)	2.5
Root Chord Length (m)	0.417
Tip Chord Length (m)	0.250
Wing Area (m ²)	0.872
Aspect Ratio	7.16
Leading Edge Sweep (°)	22.6
Quarter Chord Sweep (°)	20.4
H-stab Length	0.730
TV Flap Length (m)	0.470
Fowler Flap Length (m)	0.483
Elevator Length (m)	0.720
Rudder Length (m)	0.171
Aileron Length (m)	0.426

Table 5.2 Restraints and geometry of the aircraft.

Aircraft Restraints/Geometry	
Aircraft Empty Weight (g)	6280
Aircraft Payload Weight (g)	1280
Aircraft Maximum Weight (g)	8100
Cruise Velocity (m/s)	25
Available Static Thrust (gf)	5930
Available Dynamic Thrust at Cruise Velocity (g)	3530
Take-off Distance (m)	3.048

5.2 Airfoil Selection

5.2.1 Main Wing Airfoil Selection

The Reynolds number (Re) was the main parameter that drove airfoil selection. The predicted Re of the plane was calculated as shown in Eq. 5.1.

$$Re \equiv \frac{\bar{c}V_{\infty}}{\nu} = \frac{0.349 \cdot 25}{1.707 \cdot 10^{-5}} = 511\,131 \quad (5.1)$$

where $\bar{c}$ is the mean aerodynamic chord [m], V_{∞} is the freestream velocity [m/s], and ν is the kinematic viscosity of air [m²/s]. Using a built-in tool in XFLR5, kinematic viscosity was calculated from the average

high temperature in April (28 C) and altitude (750 m) of Tucson. The team searched the University of Illinois Urbana-Champaign database for airfoils with a maximum C_L/C_D ratio at $Re = 500\,000$. In the search parameters, the maximum thickness was set to 15% so thin airfoil theory approximations would be valid. Additionally, a minimum thickness was set to 8% so all servos and wing mounted equipment could fit internally. Maximum camber was set to 4% to minimize drag while maintaining high lift. The results returned three potential airfoils as shown in Table 5.3.

Table 5.3 Final airfoil selection.

	Max. C_L/C_D @ $Re = 500\,000$	Max. Thickness (% of chord)	Max. Camber (% of chord)
Eppler E393	124.7	11.5	4
Selig S4062	126.6	9.5	4
PSU 94-097	128.5	9.7	4

Fig. 5.2, 5.3, and 5.4 show the shapes of the three airfoils from Table 5.3.

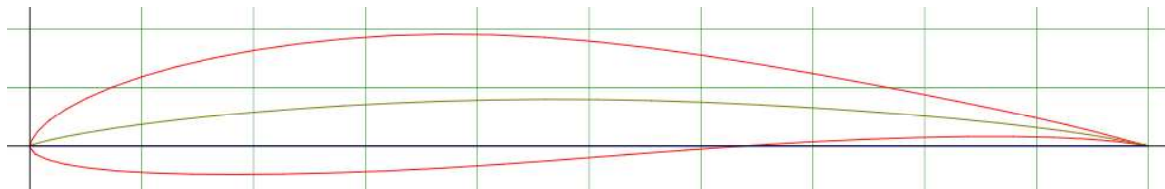


Fig. 5.1 A cross section of the Eppler E393 airfoil.

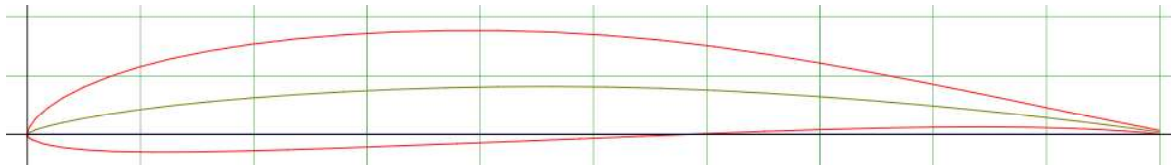


Fig. 5.2 A cross section of the PSU 94-067 airfoil.

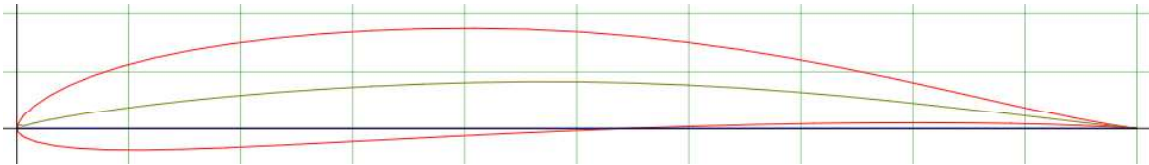


Fig. 5.3 A cross section of the Selig S4062 airfoil.

The Eppler E393 was chosen for its higher thickness, which allows more components to be embedded in the wing and increases structural integrity. Additionally, WUDBF is familiar with Eppler airfoils and has used them in the past for RC applications.

5.2.2 Stabilizer Airfoil Selection

Stabilizer airfoils are generally symmetric to ensure a symmetric and predictable C_L vs. α graph. The team decided on the NACA 0010 airfoil as it is standard for stabilizers and has been used successfully in past WUDBF projects.

5.3 Materials Selection

5.3.1 Expanded Polystyrene (EPS) Foam

Due to the prospective size of this year's aircraft, material density was a major concern. As such, EPS was proposed due to its low weight and comparative cheapness. EPS also was an optimal choice for the aircraft's wings and control surfaces due to the potential for cutting precise airfoils using CNC hotwire cutting techniques. Foam sections are also easy to join and repair, ideal for rapid prototyping and on-site maintenance.

5.3.2 Carbon Fiber (CF)

The type of carbon fiber chosen was high strength, high modulus. This was selected as the main load bearing material for the aircraft. CF was considered for use in sheets as a wing coating method, providing both strength and durability, but the high areal density of available CF weaves made this untenable. CF was selected to be the main load bearing material through the use of smaller spars running through the wing core to provide strength.

5.3.3 Fiberglass (FS) with Adhesive

1-ply weave fiberglass was selected as the wing coating material to increase the durability of the foam core and to allow for increased edge retention, especially on the extremely thin trailing edge. Two types of adhesive were tested: epoxy and polycrylic sealant. The two materials were tested by laying up foam pieces with either epoxy or polycrylic sealant to find which would produce a better cohesion between the foam and the fiberglass. The foam pieces were then covered with peel-ply and breather cloth and vacuum sealed to ensure uniform curing of the adhesive. While epoxy was slightly heavier than polycrylic, its higher toughness offset the weight consideration.

5.3.4 3D Printed Polycarbonate (PC)

Polycarbonate max was chosen to be the printer filament for the 3D printed aspects of the plane. This filament has high strength, high rigidity, and mid-range density. Another option was PETG printer filament. However, PETG has a higher density than PC, and since reducing weight was a priority, PC was chosen over PETG. PLA filament was also rejected as an option for 3D printing because PC has significantly higher tensile strength and is more durable than PLA. These qualities were desirable since the 3D printed aspects of the plane needed to be strong enough to withstand a potential crash, while also minimizing weight. 3D printing itself was chosen for certain pieces because of the ease of access to 3D printing and the ability to quickly modify or replace parts as needed. Additionally, concerns about warping in Tucson's high temperatures necessitated choosing a 3D printing material with a high glass transition temperature, of which PC's was amongst the highest available (147 °C) [20].

5.4 Structural Characteristics

5.4.1 Aircraft Frame: Foam Wings and Spar Selection

Foam was the top design choice for the body of the wings. Weight and ease of modification were primary considerations when evaluating materials. Foam is a soft, lightweight material and could easily be cut to fit the complex geometries needed to compose the wing. The foam was also receptive to the fiberglass-epoxy layup, such that the final wing design optimized strength and weight considerations.

The inclusion of carbon fiber spars allowed for the use of a foam wing under strenuous flight conditions. The maximum viable load case was determined to be 5 *g*, as that was the resultant force of the highest *g* turn the aircraft could take without stalling. This maximum load was designated as representing a factor of safety of 2, representing the minimum required factor safety for each load bearing component. Various configurations and thicknesses were tested using both Solidworks Simulation and StressCheck FEA software. Euler-Bernoulli Beam theory, along with manufacturer provided geometric data, allowed us to calculate maximum stress within the beam. Whether or not the maximum stress in any given spar would exceed the beam's rated yield stress by an unacceptable margin. Equation 5.2 allowed for the calculation of the carbon fiber tubes area moments of inertia, solely in terms of the outer and inner diameter. Since the force acting on the tube in maximum load, the cross sectional area, the value of the outer diameter, and now the area moment of inertia were known, Eq. 5.3 allowed for the maximum stress in the beam to be calculated, and allowed for the team to verify the results of the FEA analysis.

$$I = \frac{\pi(d_o^4 - d_i^4)}{64} \quad (5.2)$$

$$\sigma_{xx} = \frac{\frac{F}{A} \cdot d_o}{I} \quad (5.3)$$

Using FEA, a variety of stock carbon fiber thicknesses and diameters were tested for each spar position, with various combinations of spars in each part being tested until a suitably strong for the expected loads. The final combination chosen is recorded in Table 5.4.

Table 5.4 Spar Placement and Composition

Part	Chord position	Outer Diameter(in)	Inner Diameter (in)
Angled Wing	0.25	0.564	0.500
Angled Wing	0.5	0.564	0.500.
Central Fuselage	0.125	0.439	0.375
Central Fuselage	0.25	0.439	0.375
Central Fuselage	0.5	0.439	0.375
Boom	n/a	1.12	1.00
H Stab	0.25	0.439	0.375

5.4.2 Fuselage

The purpose of the fuselage is to provide a space for storing electronic components and serve as a mounting surface for both the radome and front landing gear. Therefore, the floor of the fuselage needs to be strong as well as light, and the shape of the fuselage needs to be aerodynamic. There are two parts of the fuselage: a carriage for holding the electronic components and taking load, and a shell to provide a protective, aerodynamic covering for the avionics. The fuselage carriage holds the batteries with 50.8 mm (2 inches) in front and back to allow the batteries free movement forward and aft to adjust for the plane CG while remaining in line with the thrust vectoring pivot point. Electronic components are located in a separate compartment, behind the batteries. Additionally, this same section contains reinforcements to mount the landing gear as well as the radome, which are thicker pieces of the fuselage. The underside of the fuselage carriage contains holes to clamp and attach to the three spars: the forward one for proper positioning, the middle one to allow clearance for the wing folding mechanism, and the rear one to prevent the fuselage from acting as a cantilever. Furthermore, parts unnecessary for structural support were removed from the floor of the carriage, so spaces were hollowed out of the floor beneath the battery.

The second component of the fuselage, the aerodynamic shell, is a casing for the carriage and is a rigid structure shell with a non-loaded covering. As the axial length of the fuselage is short and there is no load path consideration to be supported by the skin, there is structurally not an advantage to a monocoque versus a semi-monocoque construction. However, in terms of manufacturability, a rigid structure with a non-loading covering can be made the most easily. The rigid structure is composed of thin, cross-sectional bulkheads connected with longitudinal stringers, and has a thin protective covering. This design choice was selected as it saved weight and was sufficient to withstand drag forces, and is more manufacturable than a fully monocoque structure.

5.4.3 Landing Gear

Carbon fiber plates were the mounting surface for the rear landing gear due to weight and strength. This design requires the center of gravity for the aircraft be located at the thrust vectoring pivot point so the rear landing gear was positioned as far aft as possible while still remaining within the 2-foot requirement. This positioning places the landing gear 171 mm behind the center of gravity.

The front landing gear is mounted onto a coiled wire, which minimizes frontal surface area. The coiled wire will allow for ease of steering while on the ground. Additionally, this will help to dampen the force felt by the aircraft when the nose comes down on landing.

Finally, the 69.9 mm diameter wheel size was chosen to remain under the height constraint.

5.4.4 Tail Hook

The tail hook consists of a polycarbonate hook attached to a short, flexible cable, located just below the maximum deflection arc of the elevator. The hook connects to the Tail assembly through the central V-stab mounts, as these polycarbonate prints allow for a direct load path to the stiff, carbon fiber

spar at the quarter chord, bypassing the weaker foam and allowing for the operator to hold the aircraft in place during takeoff.

5.5 System and Subsystem Design

5.5.1 Drop Mechanism

The drop mechanism initially constrains the store with two rubber bands. The bands, which wrap around the middle stem of the store, are fixed by a sliding pin and two clips above the store.

The stores are loaded into the drop mechanism by first loading the rubber bands into the pin release mechanism and then sliding the pin in to constrain them. On the flight line, a team member will position the stores in the drop mechanism, pull a rubber band around each store, then hook the rubber band around a large, immobile clip on the opposite side of the bomb. This secures the bomb quickly and without the need to fit the thin pin through a tiny hole while under a time constraint.

The stores were tested to ensure success under flight conditions. With the bomb loaded into the mechanism, a mass 9 times the mass of the bomb was added to its center of gravity using washers and clamps. The rubber bands stretched heavily but did not break. The test was repeated twice, once with only the front rubber band and once with only the back rubber band. The stretching was extreme, but the bomb remained place and the additional band did not break. This confirmed the bomb would stay attached to the mechanism under an acceleration of 10g, even if one of the rubber bands broke due to fatigue. After the bombs were dropped, the rubber bands remained in their clips, which confirmed the mechanism would not drop rubber bands during flight.

There is also a variant of the standard drop mechanism design, the stacked design, which features another drop mechanism directly below it. The drop mechanism frame also includes a slanted incline between the two to ensure the top bomb falls to the side of the lower mechanism, and rolls away from the prop disk.

5.5.2 Wing Folding Mechanism

A MATLAB script was written to test the Sto-wing folding mechanism. Initially, the script calculated and visualized the position of a folded wing about an axis. The position of the axis was controlled using 2 sliders, and there was another slider to control the angle about which the wing rotated back. This initial script was intended only to visualize how a folding wing could look. Later, an additional script was written to find the optimal angle of this axis, ensuring that:

1. The folded part of the wing is parallel to the unfolded section of the wing in the unfolded configuration,
2. The leading edge of the folding section of the wing is parallel to the ground in the folded configuration to ensure the wing fits in the vertical constraint while folded,
3. The chord line of the folding wing is perpendicular to the ground, which ensures the wing fits in the horizontal constraint, and
4. The tips of the folding wings are touching the horizontal stabilizer.

It was then determined such an axis did exist that satisfies these constraints.

The script helped WUDBF determine that such a sophisticated fold could be done about only one hinge, minimizing complexity.

The complex angles required necessitated 3D printing for the hinge mechanism to accurately manufacture the precise angle for the hinge. The 3D printed part occupies the entire length of a chord, and near the 1/2 chord, holds the precisely angled hinge. The hinge itself is a sprung cabinet hinge, with an internal torsion spring. In the folded configuration, the tips of the wings face the ends of the outer vertical stabilizers, and a servo on the inside of the vertical stabilizer latches the wing in place. When time to fly, the servo releases, and the torsion spring in the hinge actuates the wings into the unfolded position. When in the unfolded position, the wing is locked into place via a pair of actuated carbon fiber spars which slide into the quarter chord of the outer folding wing mount. To allow for unimpeded rotation, and to allow for the outer folding wing mount to rest on top of the inner, various cuts and cavities were added to the mounts, allowing them to fit together in the folded configuration as shown in Figs. 5.4.1 and 5.4.2.

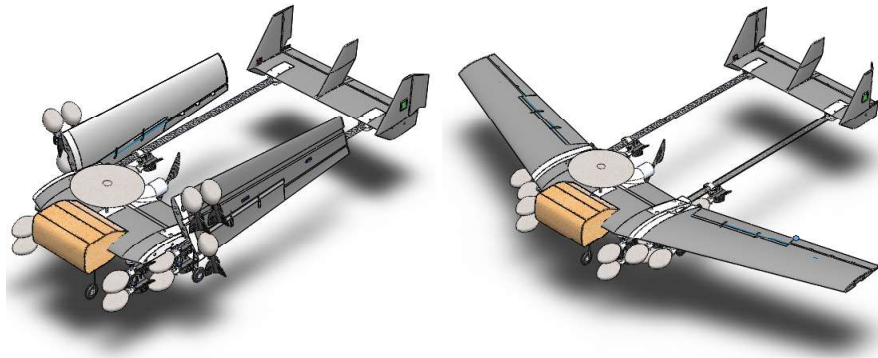


Fig. 5.4.1-2 1. (Left) Folded Configuration; 2. (Right) Unfolded Configuration.

5.5.3 Flap Mechanism

This design uses Fowler flaps, which offer superior slow-speed lift generation compared to conventional plain flaps. The Fowler flaps are connected to the main wing via off-the-shelf hinges called Robart Fowler Flap Hinges. These hinges were purchased to decrease the risk and complexity associated with flap mechanisms. Each flap has two of these hinges, which are mounted into cutouts in the flaps and main wing. Servos mounted under the wing are used to make the flaps slide out along the tracks of the hinges.

5.5.4 Thrust Vectoring Mechanism

In order to achieve the required short takeoff capabilities, the ability to vector the thrust of the propeller downwards is critical. This ability is provided by the attachment of the motor to a central flap. During takeoff, the flap will be angled downwards by a constant angle which provides an increase in C_L and a component of thrust in the vertical direction. However, this also the forward component of thrust, which decreases the forward acceleration. This lower acceleration results in a lower speed at the end of

the ramp, correspondingly less lift. Therefore, there is some optimal thrust vectoring angle between 0° and 90° . This angle was determined to be 30° by iteratively running the final configuration's struct through KM Sim, and checking which angle resulted in a launch with the highest upwards velocity immediately after the vehicle leaves the ramp.

Given the complex geometry required to merge the shape of an airfoil and the shape of a motor mount, 3D printing was selected to manufacture the center of the thrust vectoring flap. The outside edges of the flap are hot wire cut sections of EPS, which are significantly lighter than the central printed section. After printing, the central flap adheres to the outer flaps with epoxy, and then heat set threaded inserts are set into the plastic. The motor is then mounted to these inserts using its stock mounting hardware.

In order to accurately the thrust vectoring flap, a servo is mounted to the fuselage central carriage to act as a control rod. This control rod connects to a rib on the 3D printed central section that acts as a control horn. The foam outer sections of the flap are attached to the foam inner section of the wing via Du-Bro hinges. The pilot can toggle the central thrust vectoring flap down or up during takeoff.

5.5.5 Servo Selection

Servo selection was made by estimating the torques on each control surface and finding the lightest servos that could output that torque with a factor of safety of 1.5. Torques for control surfaces were computed with an online servo torque calculator [21], and torques on aircraft mechanisms were estimated from tests with servos with known torques. The chosen servos for each control surface or mechanism are outlined in Table 5.5.

Table 5.5 Chosen servos and quantity on each mechanism.

Part	Servo	Quantity
Ailerons	DS929HV	2
Rudder	Corona CS-239HV	2
Elevator	DS929HV	1
Release Servo	BMS-101HV	2
Lock Servo	DS929HV	2
Bomb Servo	BMS-101HV	16
Thrust Vectoring Servo	BMS-A53M	1
Flap Servo	BMS-A53M	2
Nose Gear Servo	DS929HV	1
Radome Servo	Feetech FS90R	1

5.5.6 Battery Cell Comparison

Because battery pack weight severely impacts the *Spirit's* lifting capabilities and range, a study was conducted to determine the lightest battery pack that fits the propulsion requirements. Such requirements (see Section 4.6.4) are listed in Table 5.6.

Table 5.6 Battery pack requirements.

Continuous Electric Power [W]	2000
Voltage, 22S NiMH [V]	26.4
Charge Capacity [Ah]	9.1-11

The lightest battery pack has the greatest specific energy, or energy per unit mass. Power output requirements provide a constraint on discharge rate. Power output and voltage define the discharge rate: the required discharge rate of the battery can be calculated from Eqs. 5.4 and 5.5:

$$Current = \frac{Power}{Voltage} = 75.8A \quad (5.4)$$

$$Continuous Discharge Rate = \frac{Current Rating}{Capacity} = 7.6C \quad (5.5)$$

This required continuous discharge rate is fairly high for NiMH batteries; a high power per unit mass, or specific power, was therefore desired.

A number of different NiMH cells were found; their specific energies, specific powers, and charge capacities are plotted in Fig. 5.5. Three cells with especially good characteristics were then compared: the Elite 2.1 Ah, Turnigy 5 Ah, and Elite 5 Ah. The Elite 5 Ah cells had the most desirable specific energy. A battery pack made out of these cells would be 123 gf lighter than a pack made out of the second-highest specific energy cells. However, the Elite 5Ah's also have a lower power density than the Turnigy 5Ah's: they only have a 6C continuous current rating.

The Elite 2.1Ah cells were also considered due to their high power density. However, these cells have a lower energy density, which would increase propulsion package mass. Furthermore, cells with charge capacities lower than 4.5Ah would need to be put into a 3P, 4P, or greater parallel configuration. More cells would be required to make a battery pack with the same energy as one made with cells of a larger charge capacity. This increases complexity, time spent making the battery pack, and the number of failure points. A larger charge capacity was therefore desired.

Due to their high specific energy, good continuous discharge rates of up to 10C, and high charge capacity, the Turnigy 5Ah cells were chosen for the battery pack. These cells only require a 2P parallel arrangement to achieve the desired charge capacity, and result in a 22S2P battery pack weighing 3.08 kgf, lower than the expected 3.2 kgf.

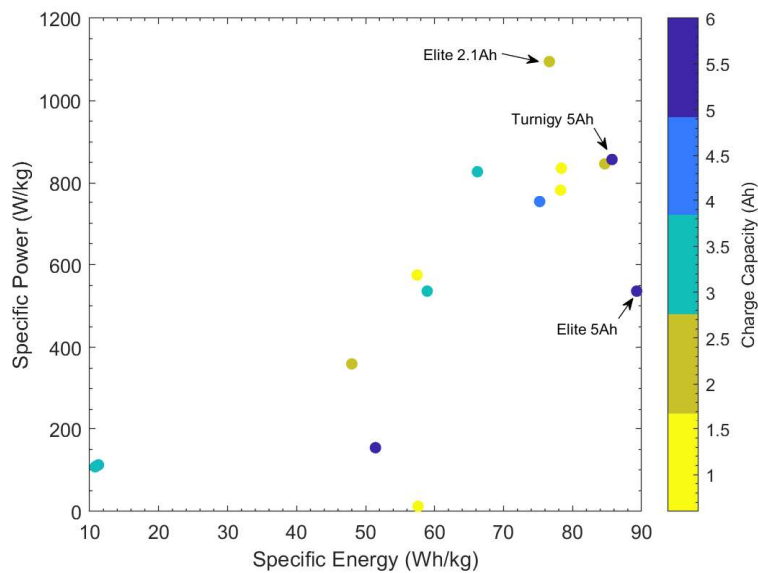


Fig. 5.5 Plot of power, energy per unit mass, and charge capacity for different NiMH cells. The three compared cells are annotated.

5.5.7 ESC Choice

The electronic speed controller (ESC) was chosen to fit within the power specifications but minimize mass. Most high power ESCs, ESCs meant for an 8S LiPo configuration at above 60 A continuous current, come with a large, heavy heatsink that adds unnecessary mass. To save weight, the team chose to attach its own aluminum heatsink with thermal paste and silicone adhesive. The Castle Edge Lite 8S 100A ESC was therefore chosen.

5.5.8 Receiver Battery Selection

The aircraft will use control bomb drops, thrust vectoring and control surfaces with a total of 30 servos. To calculate the requirements of the receiver pack, a ~10 A continuous current was initially estimated. This estimation was later verified by testing the stall and rest currents of each servo used on the plane, detailed in Section 8.2.2, and from these currents, the required energy to power the servos for the flight was calculated.

The servos will be supplied power through a UBEC regulator to provide a constant voltage of 7.0-7.4 V. The UBEC requires a minimum of 8.7 V input to provide 7.0 V output (the minimum Arduino micro input voltage), putting a lower bound on the number of cells required to provide the input. The design uses NiMH batteries, which have a discharged voltage of 1.1 V/cell, so it will need 8 NiMH cells in series to achieve greater than 8.7 V during its full lifetime. The consistent current draw from the servos was calculated to be 3.43 A with a peak draw of 10.15 A if all servos are simultaneously stalled. Using the average current draw and a flight time of 10 minutes, the necessary energy to power the servos for a

flight was calculated to be 0.44 Ah for an 8s battery pack. Applying a factor of safety of 1.5, the minimum capacity of the pack was determined to be 0.66 Ah.

However, the discharge rate was actually found to be the limiting factor of cell capacity. Although two smaller cells could have been put in parallel, the mass of all cells small enough to require a parallel connection was found to be more than double the mass of a larger, single cell. Therefore, the team decided to use two, larger cell, Turnigy 4s 1500mAh high power battery packs in series to meet the design requirements. Each pack has a weight of 0.092 kgf, for a total weight of 0.184 kgf.

5.5.9 Microcontroller Development

The main purpose of using a microcontroller, shown in Fig. 5.6 and Fig. 5.7, is to control many servos with a limited number of inputs from the transmitter. There are 30 servos to be controlled on the plane, of which 12 are control surface servos which are controlled directly by the receiver and occupy 8 channels. That leaves 18 servos to be controlled over 3 channels, which requires multiplexing those channels. An Arduino Micro was chosen to perform this task because the alternative was using multiple receivers, which would increase complexity on the transmitter side. Using the Arduino also allowed for more flexibility in how the servos are triggered, specifically in the trimming of the drop mechanism. This design utilizes the SBUS protocol to communicate over the serial port with the receiver, which can output 16 channels of data. Only one additional channel is actually used for servo drops, while two are passthrough channels used for controlling the radome and another servo should the need arise.

To condense the wiring requirements for the control and payload drop servos, a PCB was designed consisting of nine input ports and 26 output ports. Eight of the inputs and outputs are used to actuate the control surfaces of the plane. The eight outputs from the plane's receiver are routed directly to the eight output ports on the right-hand side of the board. This was done so the control servos could draw power directly from the receiver pack instead of having to pull all of its current through the receiver itself. This allows us to use high power servos and protects the receiver from over-currents. These eight outputs are then wired to multiple servos through the use of Y-connectors from each signal as needed.

The remaining 18 output signals are driven by the onboard Arduino Micro. These signals go to the payload drop servos to be used for Mission 3, and the PCB features a slider switch to toggle an output to a lower voltage setting for control of the radome servo for Mission 2.

components. The tip of the nose, was taken to be the origin, and a mutually orthogonal coordinate system was established, with the positive X-axis pointing out from the nose of the aircraft, positive Z-axis extending below the aircraft, and the positive Y-axis directed out of the right wing, relative to the nose facing forward. The masses and absolute coordinates of the aircraft's components are listed in Table 5.7.

Table 5.7 Mass and CG table. All component CGs are located in the XZ-plane.

Mission 1	Mass (kg)	X (m)	Z (m)
M1 (empty)	6.5	-0.390	0.0
Straight Wing Assembly	0.206	-0.336	-0.006
Angled Wing Assembly	0.104	-0.443	-0.003
Fuselage	0.195	-0.139	-0.009
VStab Assembly	0.135	-1.73	-0.019
Boom Assembly	0.982	-0.888	0.040
HStab Assembly	0.093	-1.72	0.032
Landing Gear	0.064	-0.466	0.173
Electronic Components	0.366	-0.089	-0.003
Main Battery	3.00	-0.079	-0.003
M2	6.7	-0.390	0.0
Radome	0.039	-0.284	-0.147
Main Battery	3.00	-0.110	-0.003
M3	8.1	-0.390	0.0
Bomb Drop Assembly	1.37	-0.319	0.108
Main Battery	3.00	-0.100	-0.003

5.7 Flight and Mission Performance

Table 5.8 provides the performance data from

The initial input parameters for the KM Sim of the *Spirit* are in Table 5.8, see figs. 5.7.1 and 5.7.2 for CL and CL/CD data.

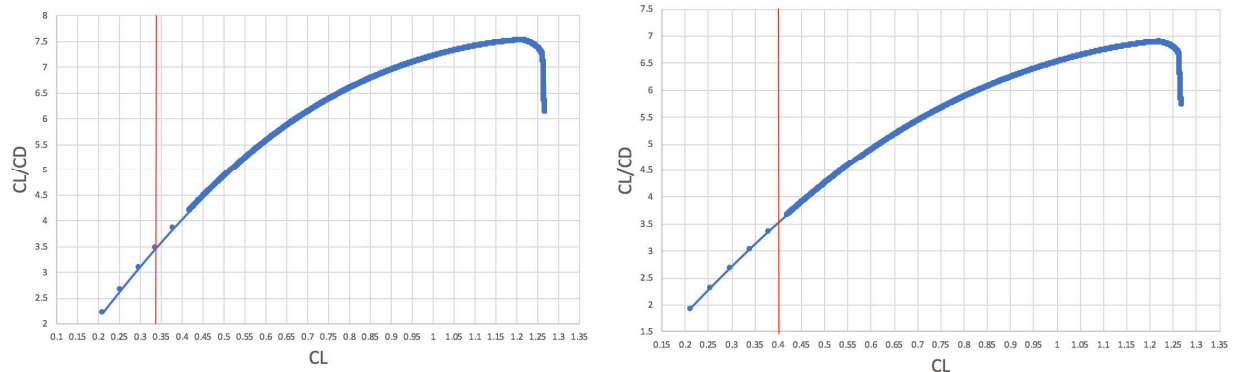


Fig. 5.7.1-2 1. CL vs. CL/CD for Missions 1 and 2; 2. CL vs CL/CD for Mission 3.

Table 5.8 Initial parameters for the *Spirit* KM Sim.

Flight Parameters	M1	M2	M3
CL Max (No Flaps)	1.25	1.25	1.25
Cl Cruise	0.34	0.34	0.40
CD	0.0794	0.0780	0.0942
L/D Takeoff	4.35	4.31	4.2
L/D Max	7.5	7.5	6.9
L/D Cruise	3.5	3.5	3.5
Vcruise [m/s]	23	33	25
Vstall [m/s]	5.2	5.3	5.5
Takeoff Gross Weight [kg]	6.8	6.8	8.1

Mission performance data for the *Spirit* according to the KM Sim are given in Table 5.9.

Table 5.9 The *Spirit* kinematic simulation performance.

Mission Parameters	M1	M2	M3	Gd
M1 Completed Successfully	Yes	N/A	N/A	N/A
Predicted Lap Time[s]	45	32.5	37.5	N/A
Number of Scoring Laps	3	3	16	N/A
Total Mission Time [s]	150	100	600	N
Score	1	1+Min_Time/ 100	18	

5.8 Drawing Package

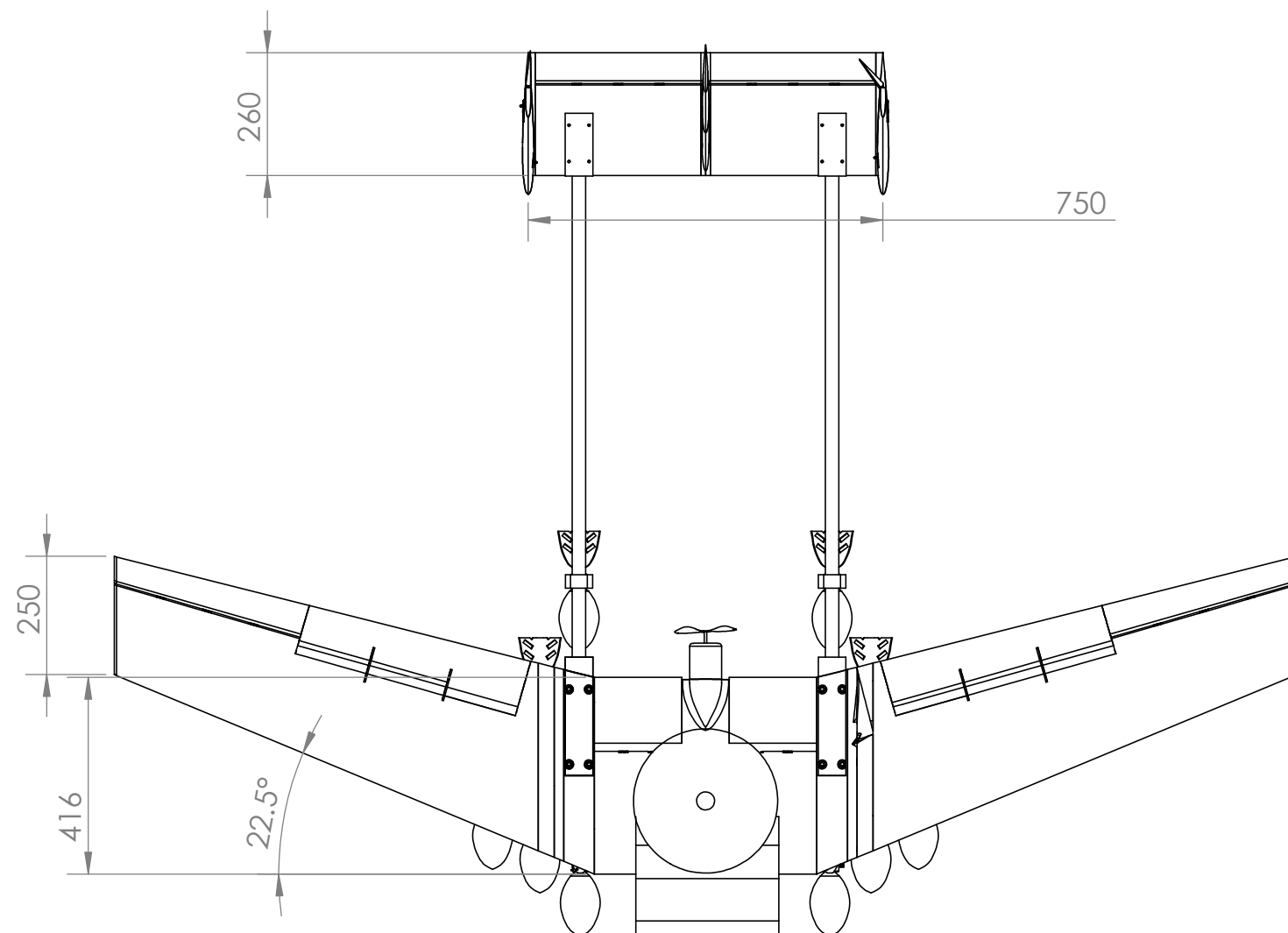
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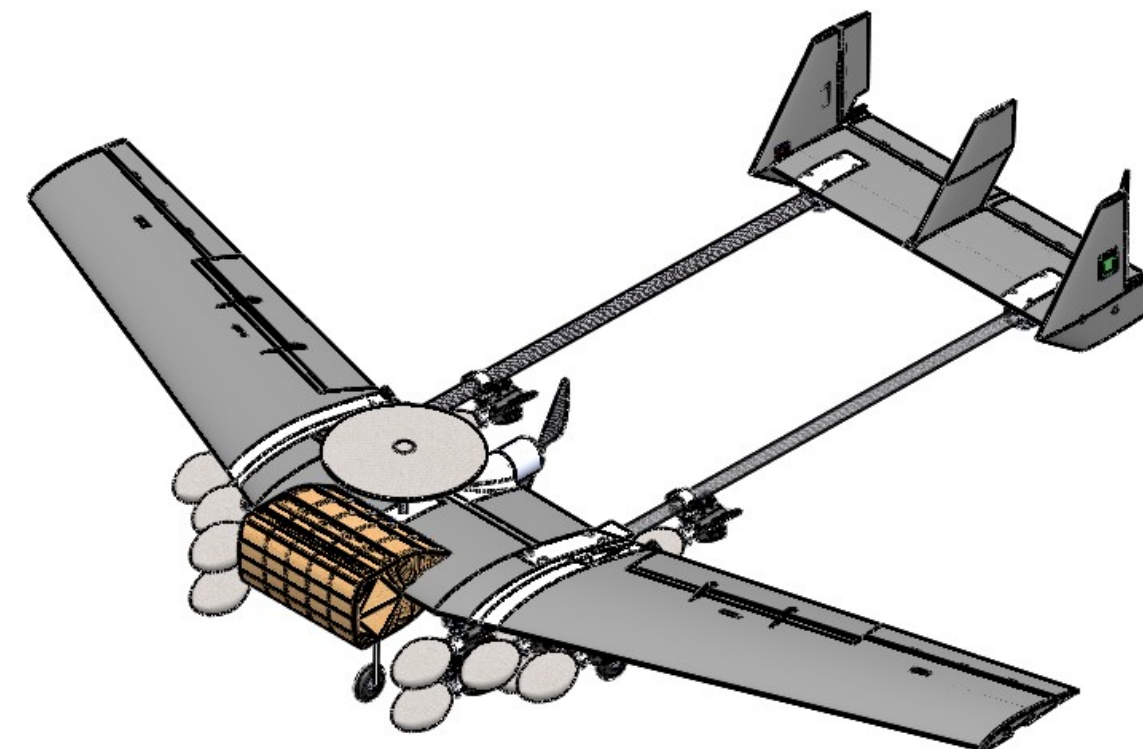
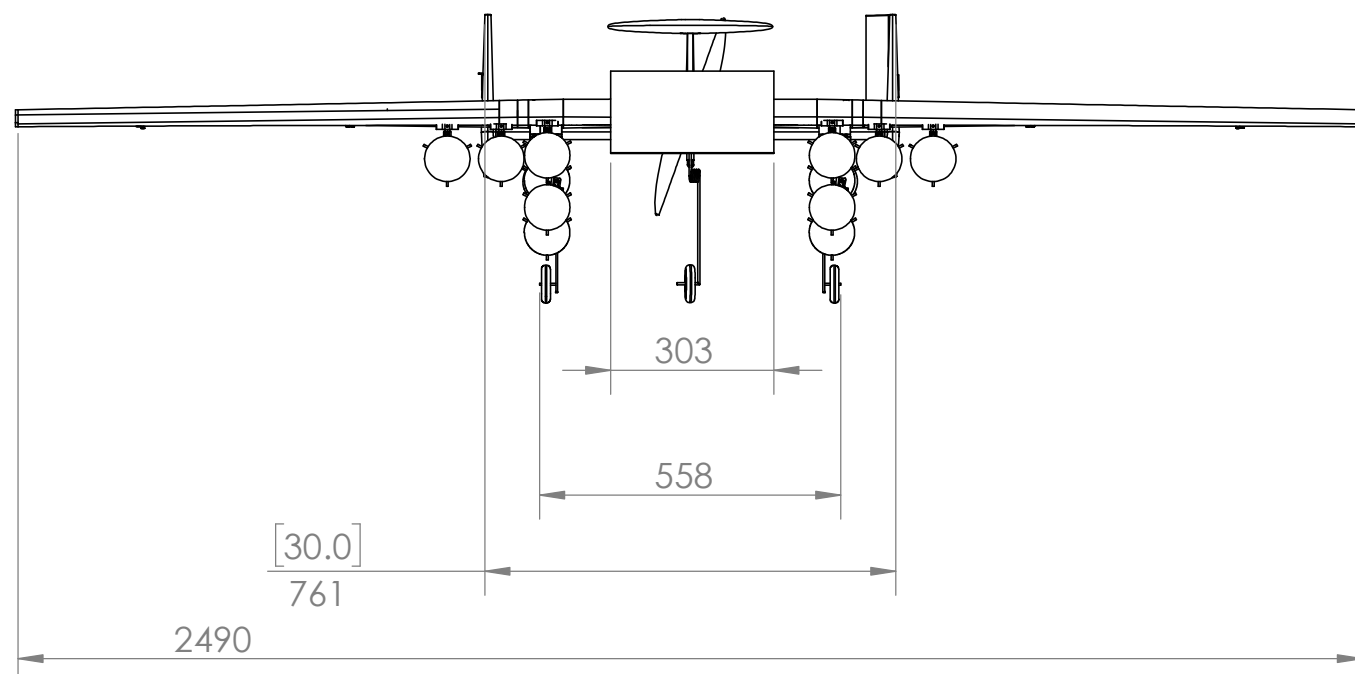
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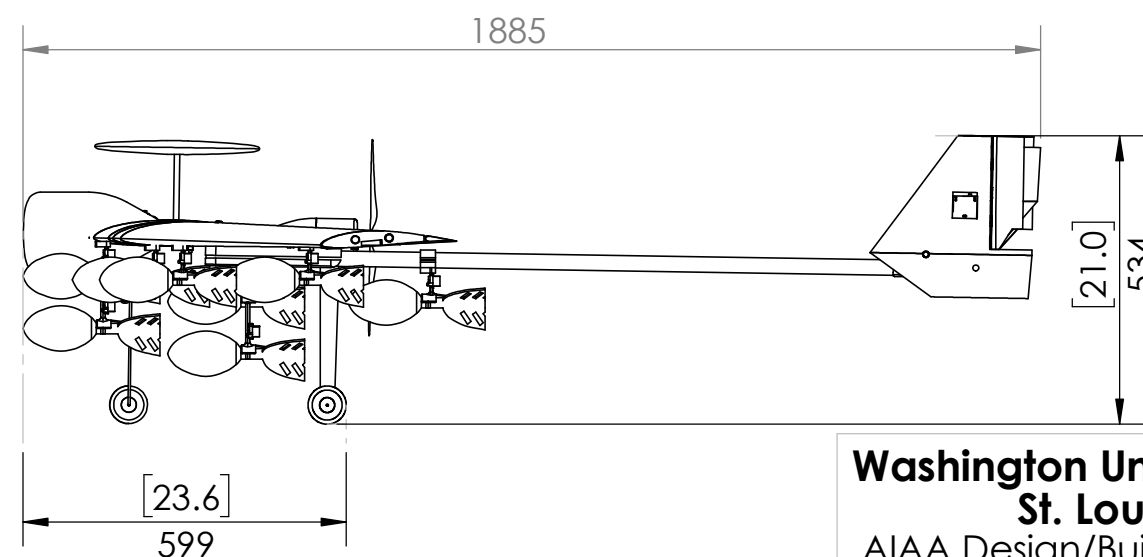
B



A



B



A

Washington University In
St. Louis
AIAA Design/Build/Fly 2019

TITLE:
Spirit of St. Louis Woodhams

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B Aircraft 3-view

SCALE: 1:14 WEIGHT: 8.1 kg SHEET 1 OF 4

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are in mm, those
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in inches

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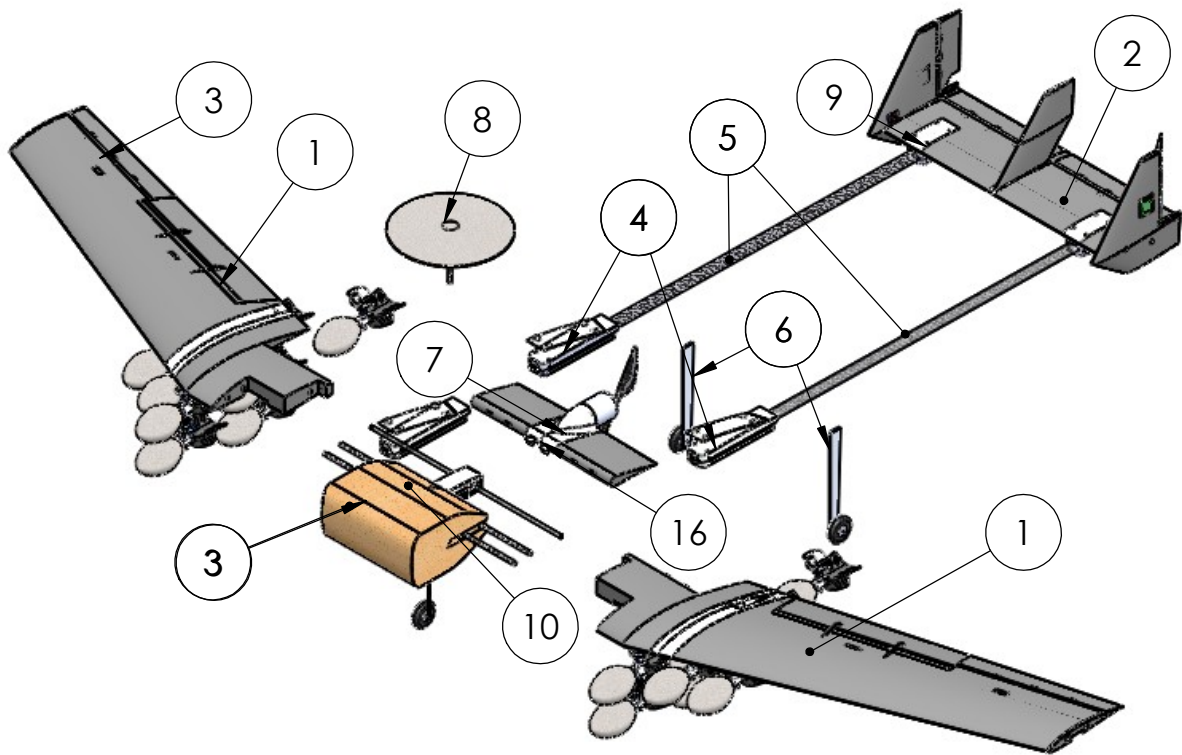
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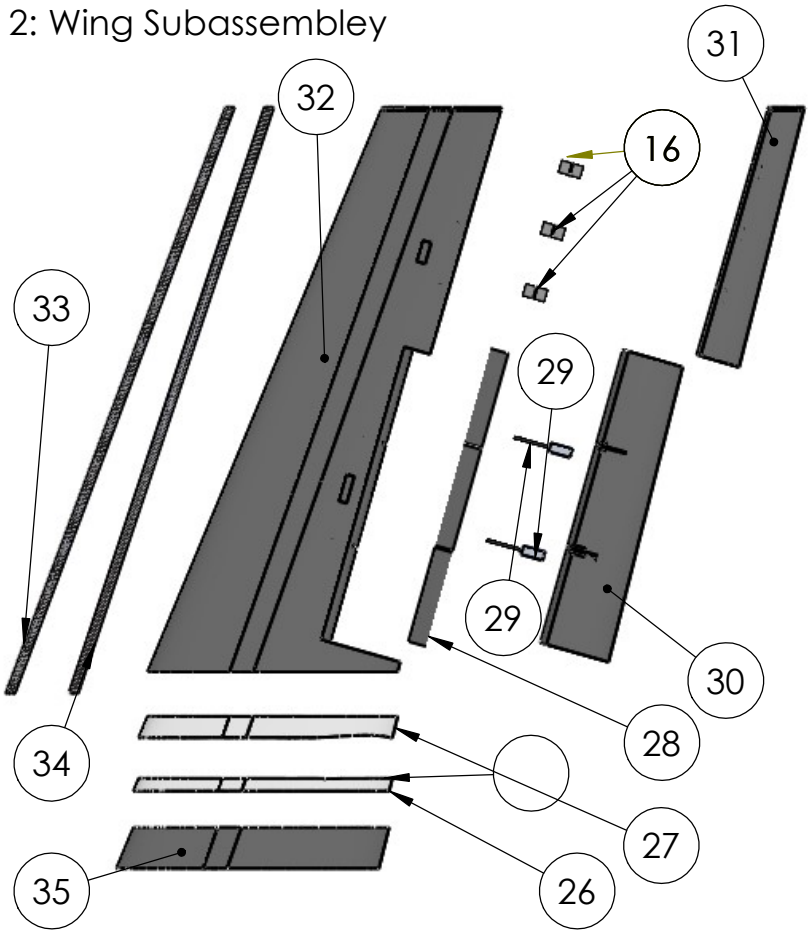
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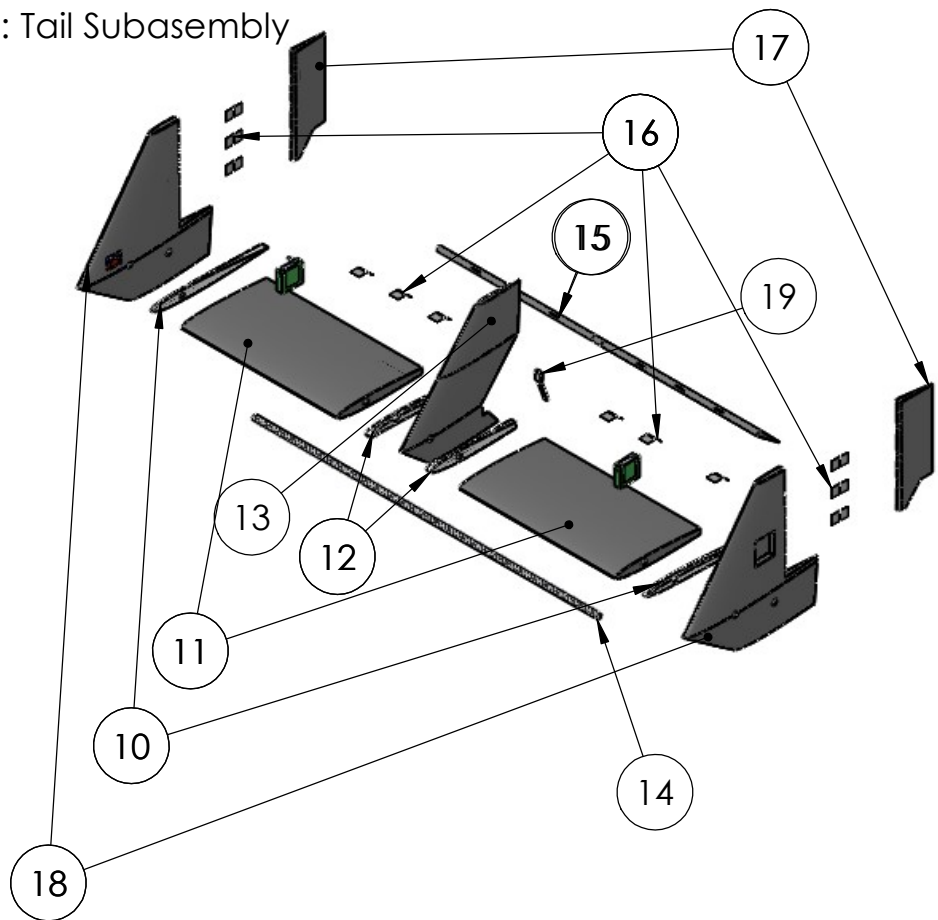
Item No.	Name	Quantity
1	Wing Subassembly	2
2	Tail Subassembly	1
3	Fuselage Subassembly	1
4	Boom Wing Clamps	2
5	Carbon Fiber Booms	2
6	Rear Landing Gear	2
7	Thrust Vectoring Mount	1
8	Radome	1
9	Boom Tail Clamps	2
10	Outer V-Stab Mount	2
11	H-Stab Core	2
12	Inner V-Stab Mount	2
13	Central Vstab	1
14	H-Stab Spar	1
15	Elevator	1
16	Dubro Hinge	20
17	Rudder	2
18	Outer V-Stab	2
19	Arresting Hook	1
20	Fuselage Central Carriage	1
21	Front Landing Gear	1
22	1/8 Chord Central Spar	1
23	1/4 Chord Central Spar	2
24	1/2 Chord Central Spar	1
25	Fuselage Outer Nacelle	1
26	Inner Folding Wing Mount	2
27	Outer Folding Wing Mount	2
28	Fowler Flap Cover	2
29	Fowler Flap Hinge	2
30	Fowler Flap	2
31	Aileron	2
32	Angled Wing Foam Core	2
33	1/4 Chord Swept Spar	2
34	1/2 Chord Swept Spar	2
35	Inner Swept Wing Foam Joint	2



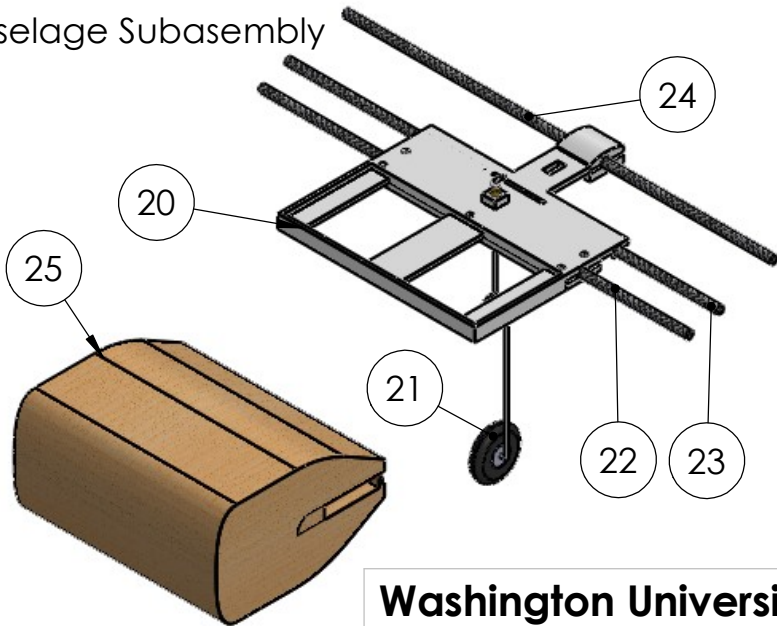
2: Wing Subassembly



3: Tail Subassembly



4: Fuselage Subassembly



Washington University In St. Louis
AIAA Design/Build/Fly 2019

TITLE:
Spirit of St. Louis Woodhams

DRAWN BY Jonathan DelGizzi
COMMENTS

SIZE	DWG. NO.	REV
B	Structural Arrangement Diagram	
SCALE: 1:18 WEIGHT: 8.1 kg		SHEET 2 OF 4

3

2

1

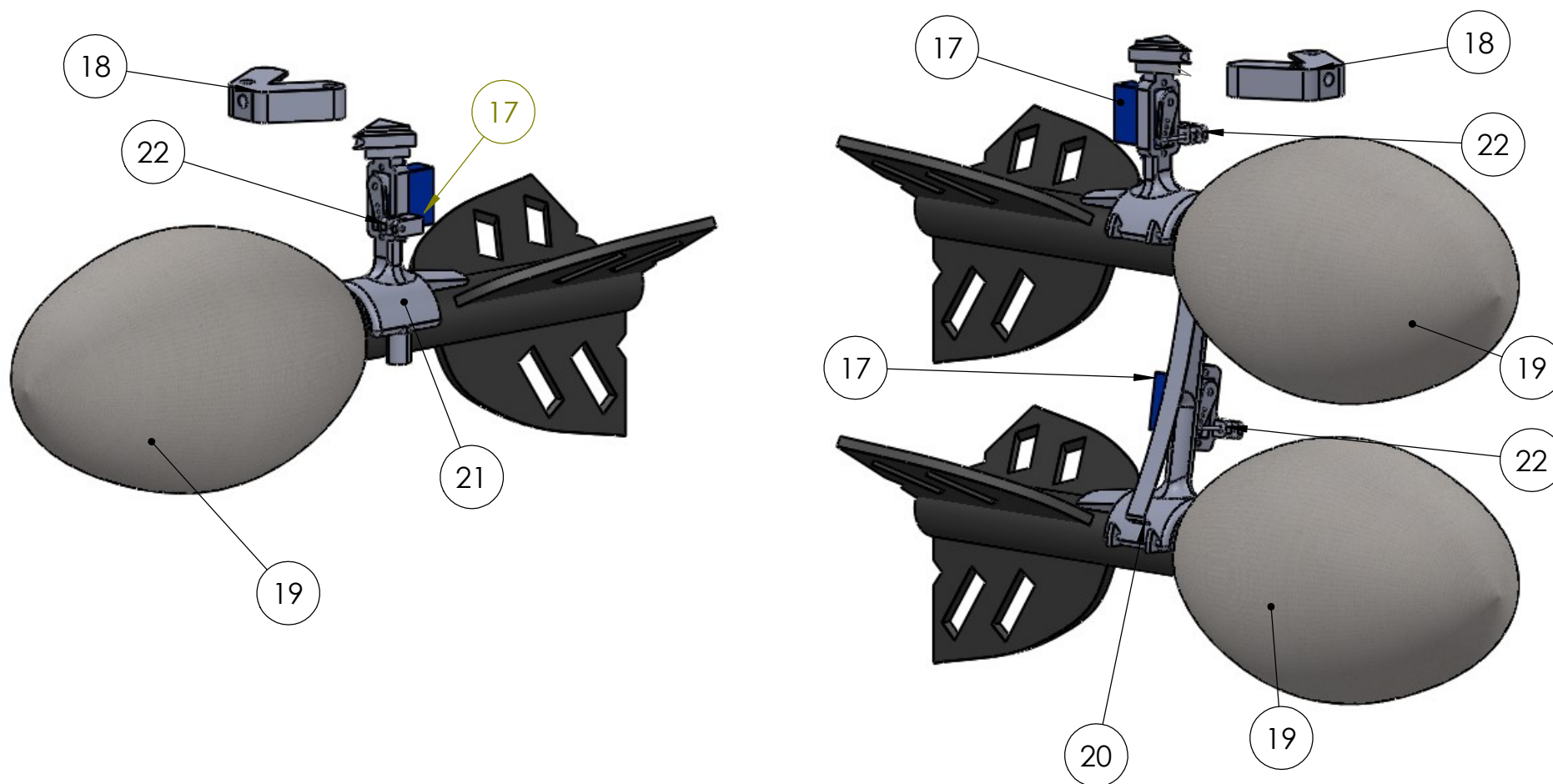
4

3

2

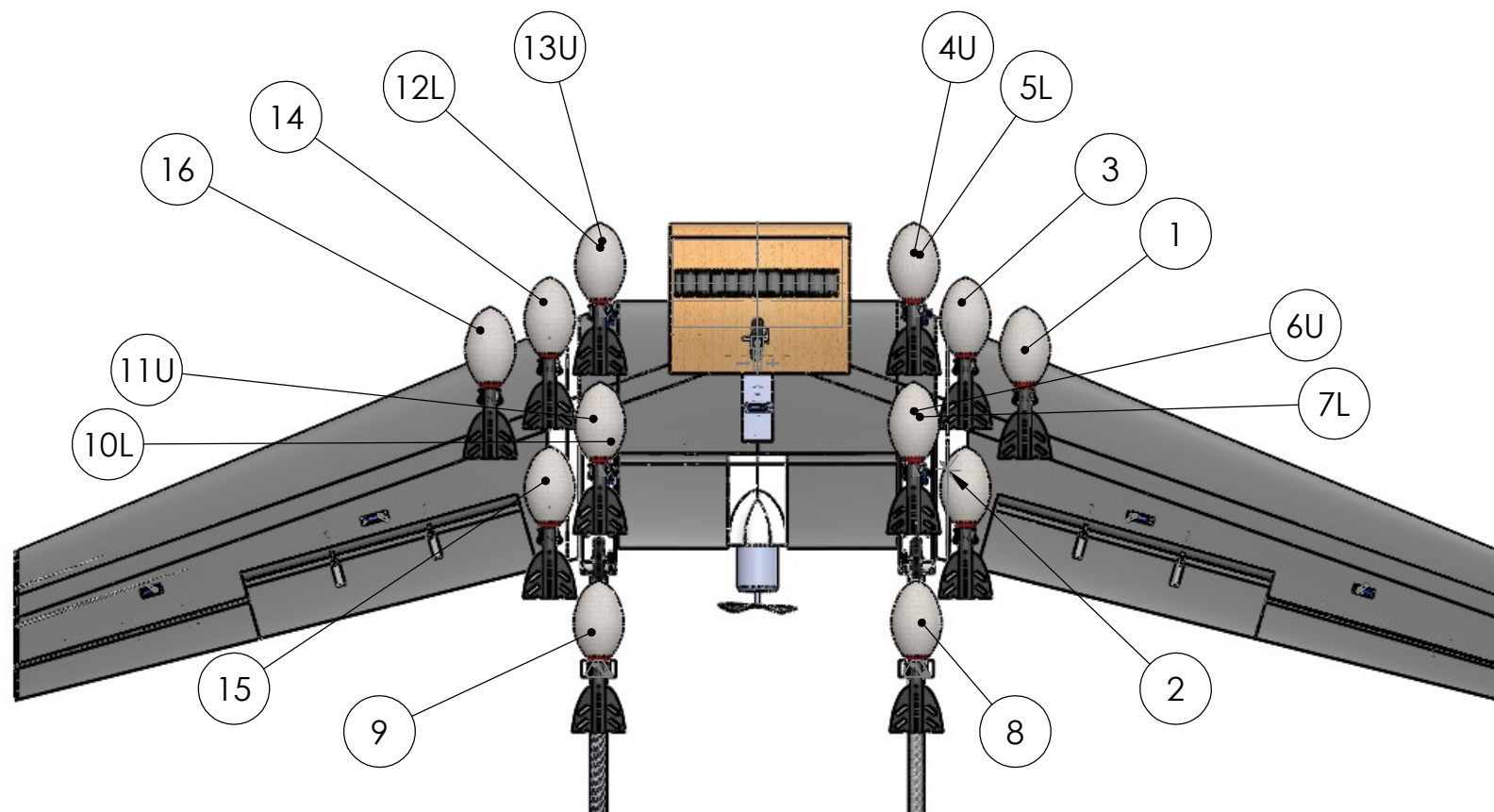
1

B



B

A



A

Part/Bomb No.	Name	Quantity
1	Attack Store 1	1
2	Attack Store 2	1
3	Attack Store 3	1
4	Attack Store 4	1
5	Attack Store 5	1
6	Attack Store 6	1
7	Attack Store 7	1
8	Attack Store 8	1
9	Attack Store 9	1
10	Attack Store 10	1
11	Attack Store 11	1
12	Attack Store 12	1
13	Attack Store 13	1
14	Attack Store 14	1
15	Attack Store 15	1
16	Attack Store 16	1
17	BMS-101HV Servo	16
18	Wing-Bomb Release Connection Mount	16
19	Attack Store Double Store Release Mechanism	16
20	Single Store Release Mechanism	4
21	Store Release Pin	8
22	Store Release Pin	16

Washington University In
St. Louis
AIAA Design/Build/Fly 2019

TITLE:
Spirit of St. Louis Woodhams

DRAWN BY Jonathan DelGizzi

COMMENTS
#U refers to the upper bomb closer to the wing. #L refers to the lower bomb

SIZE DWG. NO. REV
B Payload Acomodation Drawing

SCALE: 1:24 WEIGHT: 8.1 kg SHEET 4 OF 4

3

2

1

6. Manufacturing Plan

6.1 Manufacturing Methods Considered on Major Components

6.1.1 Wings and Control Surfaces

To construct a plane with the wingspan of the *Spirit*, the manufacturing methods needed to be simple as well as lightweight. Methods considered were a balsa and UltraCote construction as well as a foam core and surface composite layup. UltraCote, since it is a thin and lightweight plastic wrap that activates by heat and tightens to fit across frames, has the advantage of providing substantial weight savings while remaining sturdy. In addition to being lightweight, balsa and UltraCote is more aesthetic and also has the flexibility of design, however, this method was ruled out as it requires a large amount of skill and practice as well as laser cutting time. Therefore, the team determined the most efficient construction method considering time and ability is to have all airfoil components professionally foam cut to optimize speed and precision. The foam defines the shape, which is strengthened within by carbon fiber rods running along the length. Thinner components, such as control surfaces and other flaps, required strengthening through layups with carbon fiber or fiberglass sheets, chosen for their strength to weight ratio. Carbon fiber was considered as it is stronger than fiberglass, however, fiberglass has the added benefit of being more durable to surface wear. The selected fiberglass was 0.73 oz/yd², E-Glass, plain weave, selected for its balance of strength to weight as well as its compatibility of doing an epoxy layup with vacuum bagging. Furthermore, as an attempt to minimize weight while maintaining structure, the team experimented with various types of matrix adhesives such as epoxy vs. Minwax polycrylic, applying the adhesive to the wing versus the fiberglass, as well as the effects of vacuum bagging on structural rigidity and quality of layups. Section 8.3 shows all the manufacturing tests, and due to the results of the experiment the lightest and from this experiment, it was determined the foam core layup method that produced the most lightweight component at the desired toughness was an epoxy layup applied directly to the wing and vacuum bagging for 24 hours.

6.1.2 Fuselage

The fuselage is comprised of a carriage and a shell, for holding the battery and electronic components as well as aerodynamic purposes respectively. The carriage will be printed from polycarbonate as it is both light and strong, and 3D printing will allow for flexibility of design and rapid manufacturing. Since the shell is not taking any load, the manufacturing methods considered were optimized to be lightweight, and so it was proposed the shell be a rigid structure with an unloaded covering, where the structure would be made of balsa and the covering would be UltraCote or a 3D-printed thin shell. The 3D-printed shell would be easier to design and print, however, has the disadvantage of being heavier, weighing in at approximately 120 grams if printed from polycarbonate. For the same component, the balsa and UltraCote assembly weighed half that of the 3D print, at approximately 63 g. A first prototype of the UltraCote and balsa construction is shown in Fig. 6.1.



Fig. 6.1 The first prototype of the fuselage constructed from UltraCote and balsa.

Ribs and stringers located throughout the shell have the purpose of providing shape and will be made of $\frac{1}{8}$ in thick balsa, with the chord along the same axis of the grain. For strengthening the outer ribs, the methods considered were making them from 1/16 in plywood, 1/16 in basswood, or a double layer of balsa wood with perpendicularly aligned grains. The double balsa layer had the advantage of being light, however, the space available for laser cutting makes printing at perpendicular grain lines challenging. Both 1/16 in basswood and 1/16 in plywood components are strong, but basswood is slightly lighter.

6.1.3 Tail Attachment

The tail is attached to the fuselage via two twin booms. These booms needed to be light, strong, as well as hollow to allow wires to run through them. Methods considered were constructing the booms from foam, wood, or custom-ordering them in carbon fiber. While foam is light, when flying a pre-prepared foam RC plane with twin booms it was evident there was a lot of vibration and oscillation, so this led the team to opt for something stronger, such as wood or carbon fiber. Ultimately, carbon fiber was chosen as it was both lighter and stronger than wood and is readily made hollow.

6.2 Major Component Manufacturing Process

6.2.1 Wings

The foam cores of the wings are made from EPS, which was selected as it is light, sturdy, and is safe and non-toxic especially compared to other types of foam. Due to its durability, the wings were decided to not be laid up as this would be adding extra weight without increasing the strength significantly, and so a single layer of fiberglass tape was applied to both the leading edges and trailing edges, to protect it from dents and to reinforce the thin foam, respectively.

6.2.2 Control Surfaces

The precision-cut control surfaces were also made from EPS. As they are very thin and fragile especially along the trailing edge and had a large deflection along the length, it was evident they needed to be strengthened. Once the foam components were ordered from MohrComposites, a manufacturing company specializing in molded and custom-cut parts, and after evaluating the results of the experiments summarized in Table 8.2, the layup method done was the same method as trial #8, where epoxy was

applied directly to the surface of the elevator, rudder, aileron, and flaps, around which the fiberglass cloth was wrapped. The pieces were then vacuum-bagged and left to cure for 24 hours.

6.2.3 Tail

Similar to the wings, the tail and its control surfaces are also made from EPS. The control surfaces located on the tail were laid up according to the best method selected from Table 8.2, trial #8. Parts of the tail that were not control surfaces were strengthened by a single layer of fiberglass tape applied to the leading and trailing edges. Due to the shape of the airfoil of the tail, it was difficult to attach pieces of the tail together without there being gaps. To resolve this, customized pieces 3D-printed from polycarbonate allowed for smoother attachment points, so parts of the tail could be joined together more smoothly.

6.2.4 Fuselage

After experimentation with both a semi-monocoque structure and a 3D printed shell, the UltraCote and balsa design was decided upon as more beneficial because it is lighter. The inner ribs and spars are made from balsa, and the outer ribs are decided to be made from 1/16" basswood as it is lighter than plywood and strong enough to provide shape.

6.2.5 Landing Gear

The nose landing gear components are purchased from Du-Bro, a manufacturer specializing in creating components and tools commonly used in aerospace, automotive, and marine applications. The stem is a 4 mm heavy duty steel wire with a coiled spring; the wheels are 2.75" (69.9 mm) diameter and are created from treaded foam, selected for its lightness. The rear landing gear will use the same wheels, and the stem will be custom-ordered and made from carbon fiber, selected for lightness and strength.

6.2.6 Store Holder

The store holder is 3D printed from polycarbonate, as this provides flexibility in design and allows rapid iterations. The stores are held in place using rubber bands, and so a 3D printed surface allows for smoother interaction at the interface between the rubber band and store holder without needing to worry about catching on splinters or shearing forces if using wood or foam.

6.2.7 Radome

The radome will be constructed out of foam board and mounted on a 10 mm diameter hollow carbon fiber rod. The hollow rod is lightweight, as well as allows wiring to run from the radome to the electrical components of the plane located within the fuselage.

6.3 Manufacturing Milestones

The following manufacturing schedule was implemented to ensure all components were completed on time. Major milestones include having the wing foam core ordered, the fuselage printed, layups completed, and all components assembled and wiring implemented. While the first prototypes of components were being assembled, the current design was improved upon to place new foam orders.

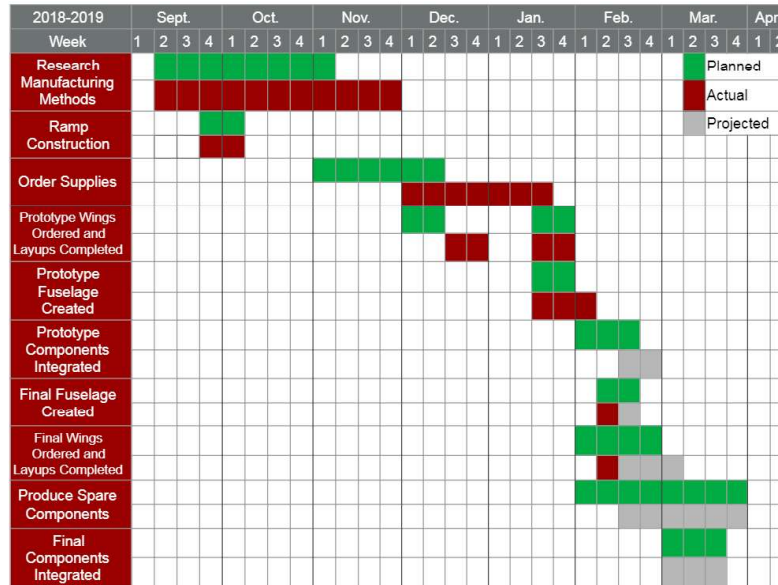


Fig. 6.2 Gantt chart of the manufacturing plan, showing actual, projected and planned schedule.

7. Testing Plan

7.1 Propulsion Testing

The objective of propulsion testing is to verify the propulsion package in Table 4.5 matches simulated results. Propulsion testing is performed with an RCbenchmark Series 1585 dynamometer. Tests conducted include battery endurance, output voltage vs. time, static thrust at 50% throttle, and torque and RPM testing to determine electrical efficiency and verify power consumption. This testing allows propeller coefficients of power and thrust to be calculated experimentally and allows static propulsion testing to be extrapolated into dynamic thrust predictions. The WUDBF testing schedule is given in Fig. 7.1, below.

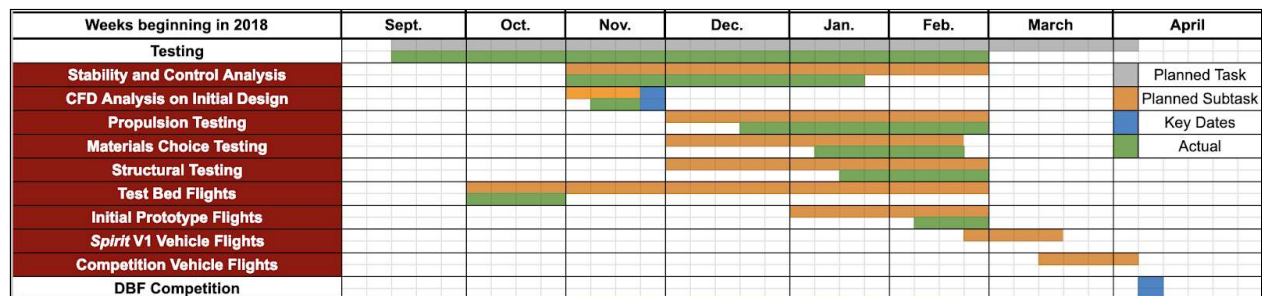


Fig. 7.1 Subsystems testing schedule.

7.2 System Testing

7.2.1 Transmitter Testing

The transmitter must be tested to ensure the aircraft enters fail-safe mode. A radio fail-safe test was performed to ensure in the case of a transmission failure, the *Spirit* would enter fail-safe mode and meet the safety requirements. The radio fail-safe mode requires the following:

1. Throttle is closed
2. The elevator is in the full up position
3. Right rudder and aileron are in the full right position
4. Flaps are in full down position

The transmitter will also be tested for an acceptable range (≥ 5000 ft is expected).

7.2.2 Servo Testing

Servos must be tested for current requirements in order to ensure that the receiver battery discharge rate, energy, and UBEC current output are sufficient, as well as to ensure that the PCB has the correct current tolerance. Servos must be tested for current at rest and while stalled to approximate real current requirements of different components on the plane. Because the team was unfamiliar with high voltage (7V) servos, the servos were also tested to ensure that their logic voltage is compatible with the microcontroller.

7.3 Manufacturing Testing

Manufacturing testing is based on determining the best manufacturing methods when considering team ability, precision capability, and maintenance of structural integrity while saving weight. For example, weight tests were conducted to compare a fuselage shell shaped by balsa bulkheads with a light covering versus a 3d-printed fuselage shell; the weight, strength, and ease of manufacturability were tested.

One important facet of foam construction is strengthening, which requires laying up foam in fiberglass. The best layup method optimized for the components to be lightweight as well as the desired toughness, so experiments of varying combinations of matrix adhesives such as epoxy vs. Minwax polycrylic, adhesive application, i.e. applying the adhesive to the wing directly vs. the fiberglass cloth, as well as the effects of vacuum bagging on structural rigidity and quality of layups were conducted. For a given experiment, the type of adhesive used, how it was applied, as well as whether or not it was vacuum-bagged, is indicated in Section 8.3.

7.4 Pre-Flight Checklist

Before each mission during the competition, a team member will read through Table 7.2 to make sure the *Spirit* is operational and safe.

Table 7.2 Flight checklist.

Component/System	Action to take
Environment	☐ Record atmospheric conditions, wind speed, and wind direction
Control surface	☐ Check command servo deflection and control surface response
Propulsion	☐ Check motor and prop direction ☐ Check thrust vectoring system ☐ Check throttle power and vibration
CG Location	☐ Confirm CG is on the thrust vectoring pivot point
Landing gear	☐ Confirm landing gear is properly attached ☐ Check wear on wheels
Batteries	☐ Check propulsion battery is securely connected, mounted and fully charged ☐ Check receiver battery is securely connected, mounted and fully charged
Mission Specific	☐ Confirm radome is properly attached (Mission 2) ☐ Confirm the radome turns (Mission 2) ☐ Confirm payload and payload drop mechanisms are properly secured (Mission 3) ☐ Ensure microcontroller is in flight mode and switches are in correct state for radome/drop
Tail Boom	☐ Confirm booms are properly attached
Safety	☐ Clear takeoff area ☐ Check motor and propeller screws/nuts are tight and in good condition ☐ Visual inspection of plane to ensure no parts are damaged and secure

8. Performance Results

8.1 Propulsion Testing Results

Thrust, torque, voltage, RPM, and current were measured over time and analyzed for a static thrust test at 50% throttle, as shown in Fig. 8.1 and Fig. 8.2. Testing was limited to 50% throttle due to the thrust limits of the dynamometer.

Results were mixed. Electrical efficiency stayed at ~90% over the lifetime of the battery, higher than the 86% efficiency expected from the km sim. This is expected since lower throttle values typically lead to a small efficiency increase. On the other hand, the NiMH cells were fully drained when 34% capacity was expected to be left in them. It is possible this lower-than-expected capacity occurred because the NiMH cells were fairly new and had only been discharged once before. New NiMH cells can take several charge-discharge cycles to reach full capacity. This could also be due to damage done to the cells during pack manufacturing, but this is considered unlikely. Spot welding, used to build the pack, is the preferred industry method for battery pack fabrication and was performed with care and caution. It is also possible some of the cells have a capacity somewhat lower than reported by the manufacturer. Elite 5000 mAh NiMH cells (see Section 5.5.6) will be considered as a replacement for these Turnigy 5000 mAh cells if further tests show no increase in capacity.

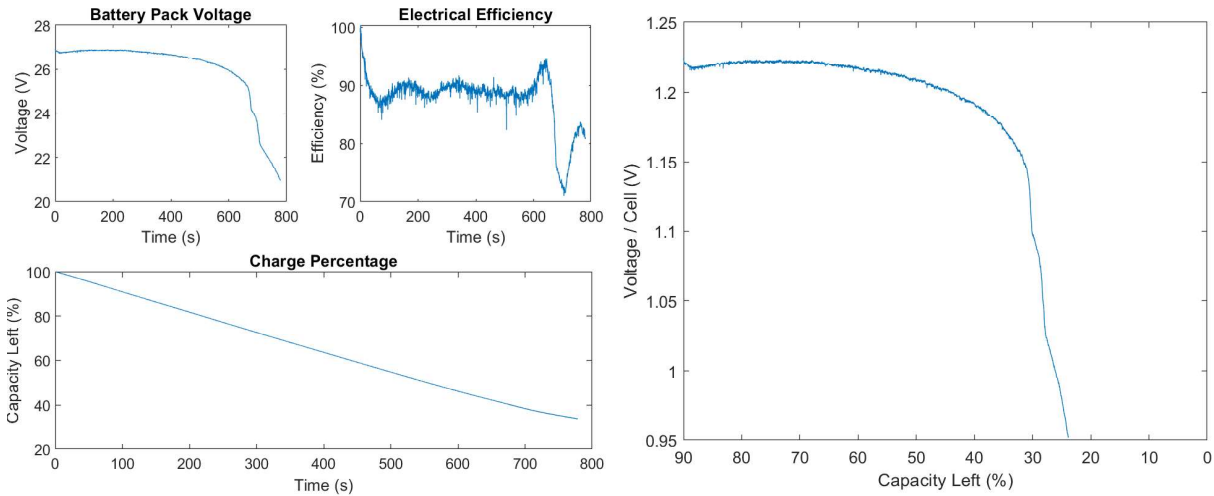


Fig. 8.1.1-2 1. Several key motor test parameters plotted over time at 50% throttle; 2. Voltage per cell plotted over percent capacity. Capacity is over 20% less than expected.

8.2 System Testing Results

8.2.1 Transmitter Testing Results

Transmitter fail-safe was tested on the flight line. The transmitter was switched off and the control surfaces performed as desired (described in Section 7.2.1), verifying fail-safe mode.

The transmitter range was also tested on an open field to determine the maximum flight distance before loss of reception and entry into fail-safe mode. Maximum transmitter range was determined to be about 1500 ft by later estimation in Google Earth. Although this distance seems reasonable for flight around the course, which only requires a 500-600 ft transmission range, it was much lower than the expected 5000 ft and settings will be checked before the next flight test to improve the range.

8.2.2 Servo Testing Results

Servo currents were tested to determine the size of the receiver pack required for a 10-minute flight. Current measurements were taken with the servo at rest and stalled, shown in Table 8.1.

The predicted current was determined by assuming all control surface servos are stalled for the duration of the flight and the payload release servos draw their rest currents since these servos only need to provide brief torques during store drop.

Table 8.2 Tested servo currents, including per-servo and overall currents.

Servo Name	# of Servos	Predicted Current (A)	Rest Current / Servo (A)	Stall Current / Servo (A)
CS-239HV	2	0.8	0.03	0.4
DS929HV	6	2.4	0.01	0.4
BMS-101HV	18	5.4	0.01	0.3
BMS-A54M	3	1.2	0.015	0.4
FS90R	1	0.35	0.02	0.35
Total on Plane	30	10.15		

The Corona DS929HV servos, used on the control surfaces, were also tested for minimum logic voltage. Because the team was unfamiliar with these higher voltage servos, which run on ~7 V, and the

logic voltage on these servos corresponds to the 5 V Arduino output, the minimum logic voltage for smooth servo operation was determined. Resistance due to the long, thin wires that must traverse over the entire length of the craft also decreases the logic voltage that reaches the servos.

To test this, the logic voltage was decreased until jitters were introduced into servo rotation. An oscilloscope was used to measure logic voltage. For smooth operation, it was found that logic voltage must be above $V_{\text{logic, min}} = 2.4 \text{ V}$. The absolute minimum for any servo response was determined to be $V_{\text{logic, abs min}} = 1.4 \text{ V}$. These voltages are more than adequate for the Arduino voltage even after the small voltage drop from the wires.

8.2.3 Drop Mechanism Testing Results

The bomb drop mechanism required some adjustment after an early test flight. Because the bombs are foam, a propeller strike was not envisioned as a catastrophic failure. The drop mechanisms were mounted near the midline of the plane to reduce negative effects from having a different number of bombs on either side of the fuselage. However, upon release, a bomb fell into the propeller disk causing the propeller to break and destroying the motor, nearly resulting in a crash. To combat this issue, the drop mechanisms have been moved further away from the midline. Also, a small guide ramp was added to the design of the mechanism to ensure the bomb will always fall away from the propeller disk.

8.3 Manufacturing Testing Results

Information about layup testing and its results are included in Table 8.2.

Table 8.2 Results from airfoil layup experiments used to determine the most lightweight layup method.

Trial #	Airfoil Layup Method			Mass (g)
	Matrix Adhesive	Application	Vacuumed Bag?	
1	Polycrylic	On wing	Yes	10.38
2	Polycrylic	On wing	No	10.89
3	Polycrylic	On wing	Yes	15.64
4	Epoxy	On fiberglass	Yes	16.02
5	Epoxy	On fiberglass	Yes	17.70
6	Epoxy	On wing	No	23.31
7	Epoxy	On wing	Yes	28.05
8	Epoxy	On wing	Yes	12.16

From these results, It appears the polyacrylic layup method produced the lightest wing. Qualitatively, however, the trailing edges of these products were significantly less stiff than products laid up with epoxy. Therefore, it was determined the foam core layup method that produced the most lightweight component at the desired toughness was trial #8, which is an epoxy layup applied directly to the wing along with vacuum bagging for 24 hours. Although trial #7 used the same methods, the end result was much heavier. But with trial #8 it can be seen the same method can be done and it is possible for a lighter wing to be produced.

8.4 Flight Testing Results

8.4.1 RC Ready Made Anaconda

The RC Ready Made Anaconda was flight tested underneath competition conditions with a ramp take off. At the time, WUDBF did not have its NiMH batteries and instead used lithium ion batteries so additional weight was added to the anaconda model to more accurately represent competition conditions. The RC Ready Made Anaconda was able to take off from the ramp with ease and with no usage of slats or flaps, but dropped in altitude after leaving the ramp. With slats or flaps it took off and did not experience any drop off the end of the ramp. This validated the KM Sim and allowed WUDBF to pursue additional aircraft analyses.

8.4.2 Tests of Prototype Spirit

Test flying the prototype of the *Spirit* yielded significant insight into the feasibility and operation of the aircraft and its mechanisms. The test flights proved the aircraft was able to complete the required ramp take off, and that avoiding rotating on take off was critical to avoid a tail strike. During flight, the aircraft required some up-trim, but was otherwise statically stable and well damped. Overall, the *Spirit* was naturally stable and flew well, maintaining good roll, yaw, and pitch rates. Initial control response was sluggish which is likely due to the mass of the aircraft. With some exponential trim this sluggish response was nullified. The store dropping mechanism was tested during a separate test flight and did not work properly; it was determined the tolerances of the mechanism needed to be adjusted, and after adjustment we were able to drop a store. Through trial and error testing with an experienced pilot it was determined the best landing approach was at a shallow angle of about 15°, however a higher angle would be necessary if a landing was attempted with a full store loadout. The aircraft handled crosswinds well and there were no observed problems with rollout.

9. Bibliography

- [1] Barnstorff, Kathy, "Ten-Engine Electric Plane Completes Successful Flight Test," NASA, 30 April 2015, <https://www.nasa.gov/langley/ten-engine-electric-plane-completes-successful-flight-test>.
- [2] "F-35B Lightning II," Lockheed Martin Corporation, <https://www.f35.com/about/variants/f35b> [retrieved 21 Feb 2019].
- [3] "Greased Lightning," Advanced Aircraft Company, <http://www.advancedaircraftcompany.com/greased-lightning/> [retrieved 21 Feb 2019].
- [4] "RMRC Anaconda - Kit," Ready Made RC, LLC., <https://www.readymaderc.com/products/details/rmrc-anaconda-kit> [retrieved 21 Feb 2019].
- [5] "Carbon-Z Cub 2.1m BNF Basic with AS3X," Horizon Hobby, <https://www.horizonhobby.com/carbon-z-cub-bnf-basic-with-as3x%C2%AE-technology-efl10450> [retrieved 21 Feb 2019].

- [6] Tegler, Eric, "One man designed and build the ultimate bush plane," *Ars Technica*, Conde Nast, 11 Aug 2018.
- [7] "The P-38: When Lightning Strikes," Lockheed Martin Corporation, <https://www.lockheedmartin.com/en-us/news/features/history/p-38.html> [retrieved 21 Feb 2019].
- [8] "2018-19 DBF Rules," 2018, AIAA.
- [9] Myers, Ken, "Wing Cube Loading, *The Ampeer*, Jan 2014, <http://www.theampeer.org/M1-outrunners/M1-outrunners.htm#CWL>.
- [10] Caughey, David A., "Introduction To Aircraft Stability and Control Course Notes for M&AE 5070," Sibley School of Mechanical & Aerospace Engineering, Cornell University, Ithaca, New York, 2011, pp. 25-101.
- [11] Foster, Tyler M., and Bowman, Jerry, "Dynamic Stability and Handling Qualities of Small Unmanned-Aerial-Vehicles," 43rd AIAA Aerospace Sciences Meeting and Exhibit, Aerospace Sciences Meetings, pp. 37-56, <https://doi.org/10.2514/6.2005-1023>.
- [12] Prandtl, L., "Applications of Modern Hydrodynamics to Aeronautics," NACA-TR-116, Jan 1923, pp. 190
- [13] Bowers, Albion H., Murillo, Oscar J., Jensen, Robert R., Eslinger, Brian, and Gelzer, Christian, "On Wings of the Minimum Induced Drag: Spanload Implications for Aircraft and Birds," NASA/TP-2016-219072.
- [14] Abbott, Ira H. and von Doenhoff, Albert E., "Theory of Wing Sections", Dover Publications, Inc., New York, 1959.
- [15] Paulson, John W., 1976, "Wind-Tunnel Investigation of a Fowler Flap and Spoiler for an Advanced General Aviation Wing," NASA TN D-8236, Washington, D.C., Jun 1976.
- [16] "A350 Aircraft Characteristics - Airport and Maintenance Planning," Airbus S.A.S., 2005.
- [17] "777-200LR / -300ER / -Freighter Airplane Characteristics for Airport Planning," Boeing Commercial Airplanes, D6-58329-2, May 2015.
- [18] McCormick, Barnes W., "Aerodynamics, Aeronautics, and Flight Mechanics," John Wiley & Sons, New York, 1979.
- [19] Drela, Mark, "First-Order DC Electric Motor Model," MIT Aero & Astro, Feb 2007.
- [20] "PolyMax PC," PolyMaker, <https://polymaker.com/product/polymax-pc/> [retrieved 22 Feb 2019].
- [21] Gadd, Chuck, "Servo Torque Calculator," Scale Flyers of Minnesota, <http://www.mnbigbirds.com/Servo%20Torque%20Calculator.htm> [retrieved 22 Feb 2019].